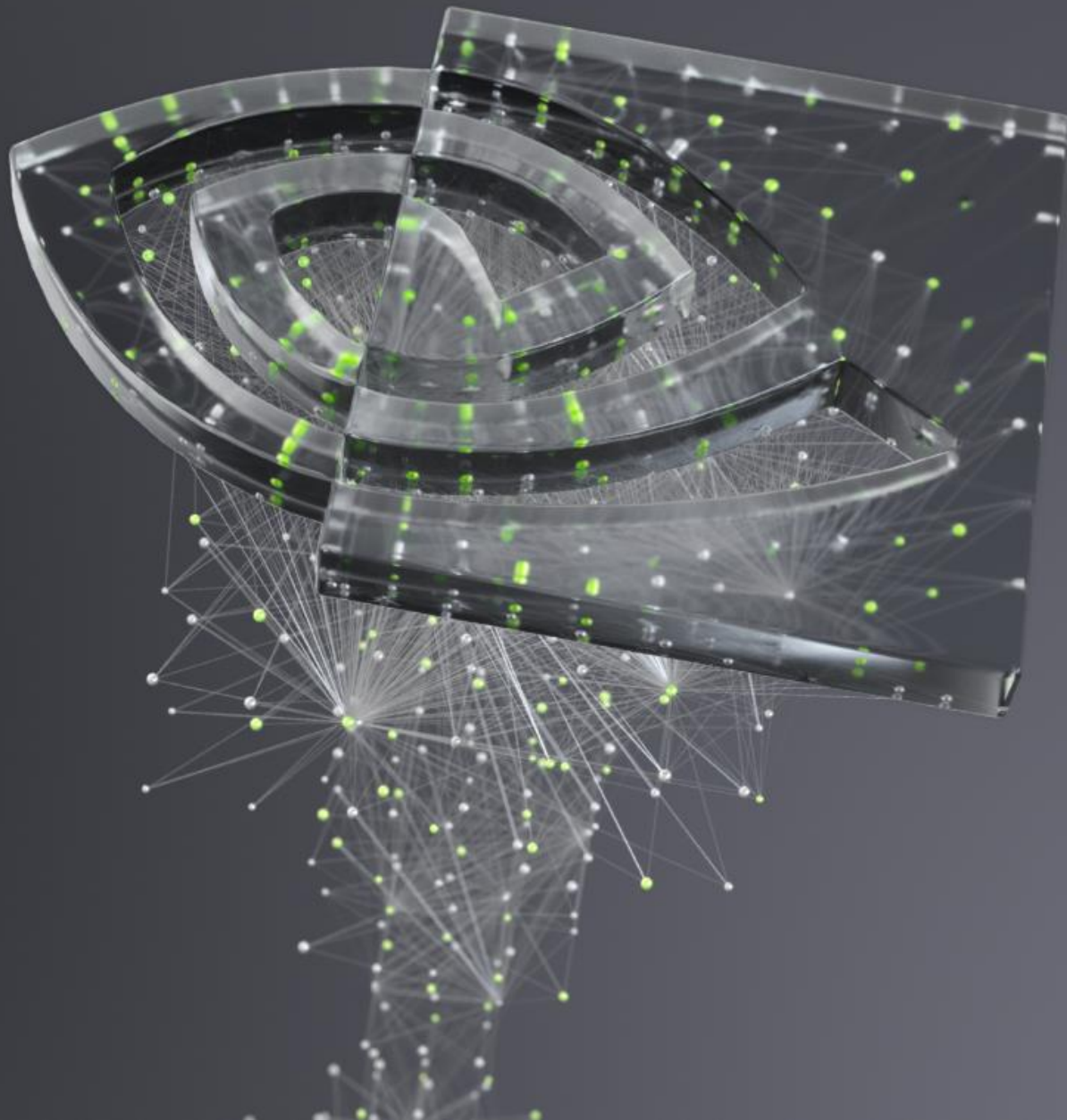




DEEP
LEARNING
INSTITUTE

딥러닝의 기초

2부: 뉴럴 네트워크의 트레이닝 방식



목차

1부: 딥러닝 소개

2부: 뉴럴 네트워크의 트레이닝 방식

3부: CNN(Convolutional Neural Network)

4부: 데이터 증강 및 배포

5부: 사전 트레이닝된 모델

6부: 고급 아키텍처

목차 – 2부

- 요약
- 더 단순한 모델
- 뉴런에서 네트워크로
- 활성화 함수
- 과적합
- 뉴런에서 분류로

핸즈온 요약

방금 어떤 것들을 배웠나요?

데이터 로드 및 시각화

데이터 편집(재구성, 정규화, 범주형)

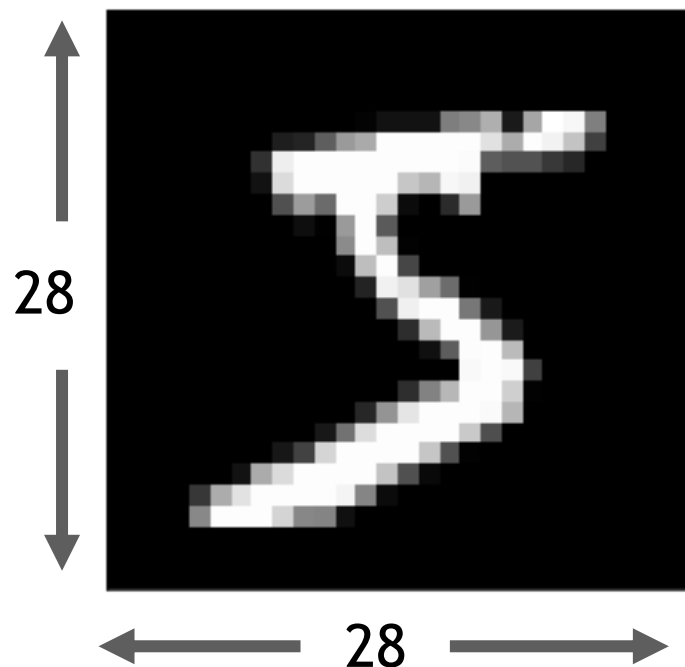
모델 생성

모델 컴파일

데이터에 대해 모델 트레이닝

데이터 준비 (DATA PREPARATION)

배열을 통한 입력



→ [0,0,0,24,75,184,185,78,32,55,0,0,0...]

데이터 준비 (DATA PREPARATION)

범주형(categorical)으로 타깃팅

0 → [1,0,0,0,0,0,0,0,0,0]

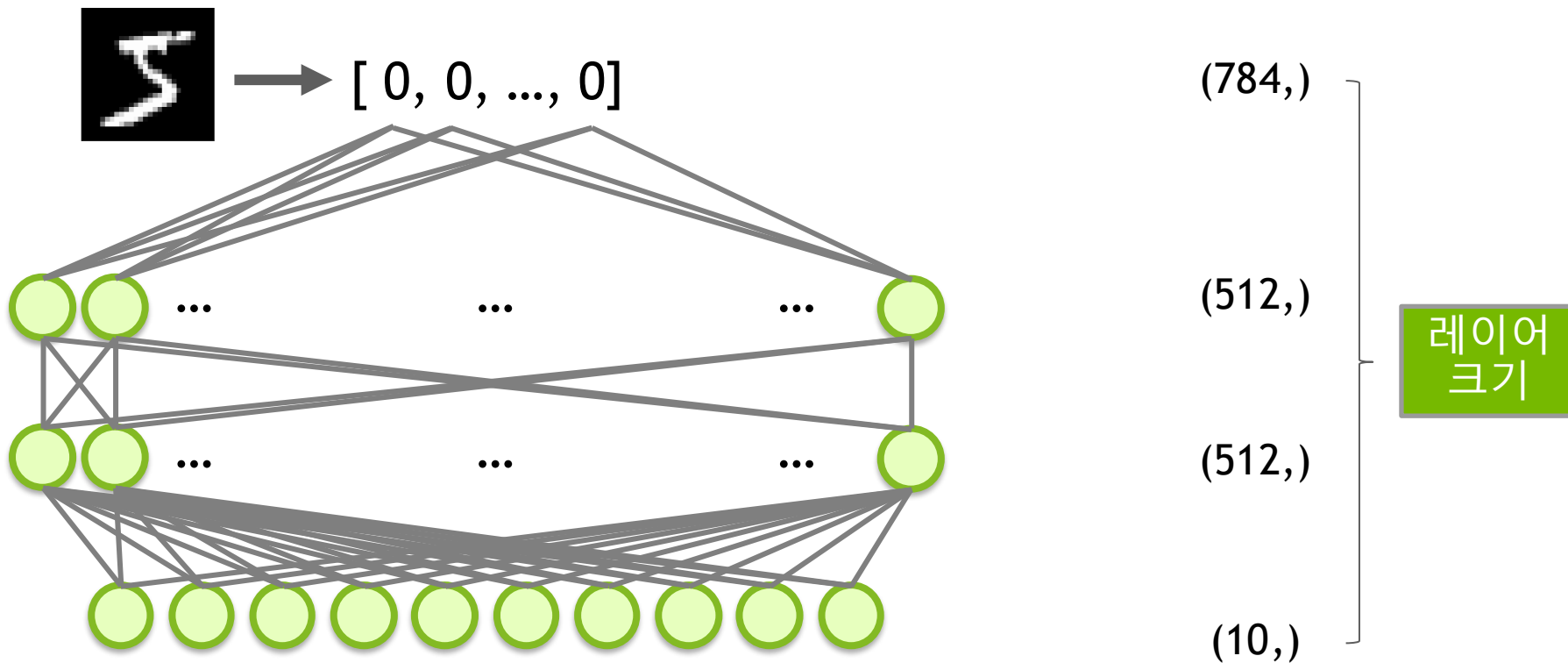
1 → [0,1,0,0,0,0,0,0,0,0]

2 → [0,0,1,0,0,0,0,0,0,0]

3 → [0,0,0,1,0,0,0,0,0,0]

•
•
•

트레이닝되지 않은 모델



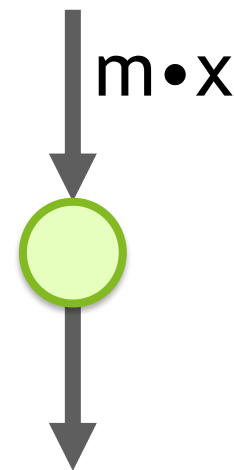
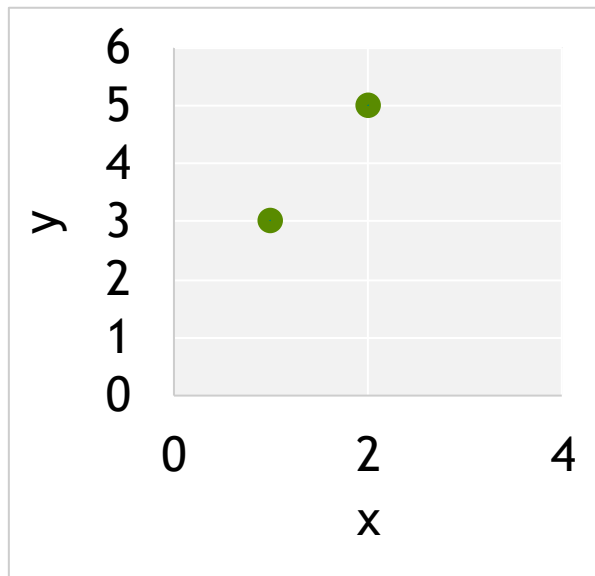


더욱 단순한 모델

더욱 단순한 모델

$$y = mx + b$$

x	y
1	3
2	5



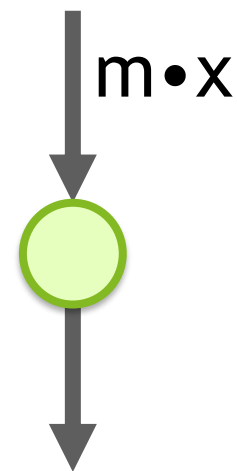
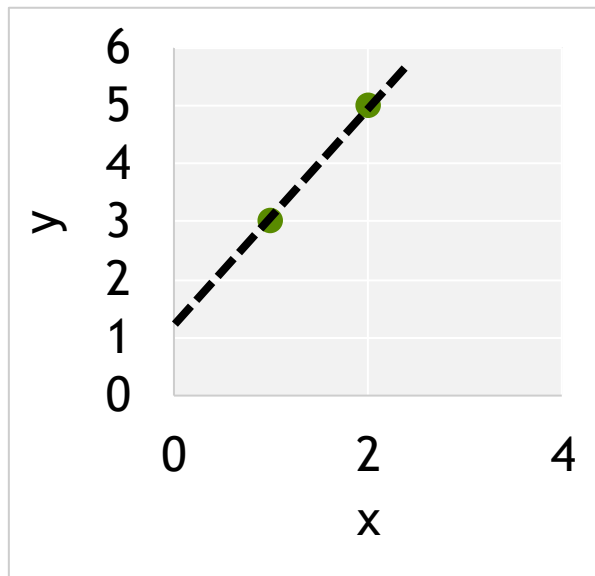
$m = ?$

$b = ?$

더욱 단순한 모델

$$y = mx + b$$

x	y
1	3
2	5



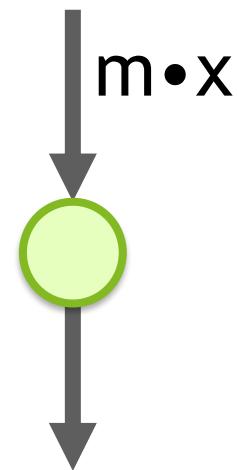
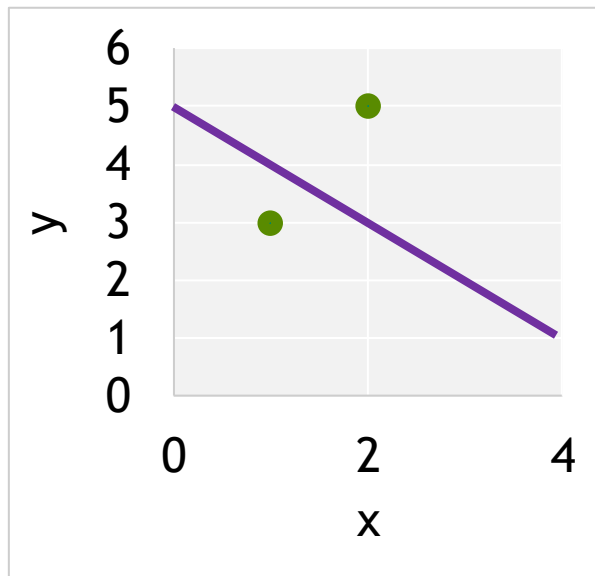
$$m = ?$$

$$b = ?$$

더욱 단순한 모델

$$y = mx + b$$

x	y	$\hat{y}$
1	3	4
2	5	3



무작위로 시작

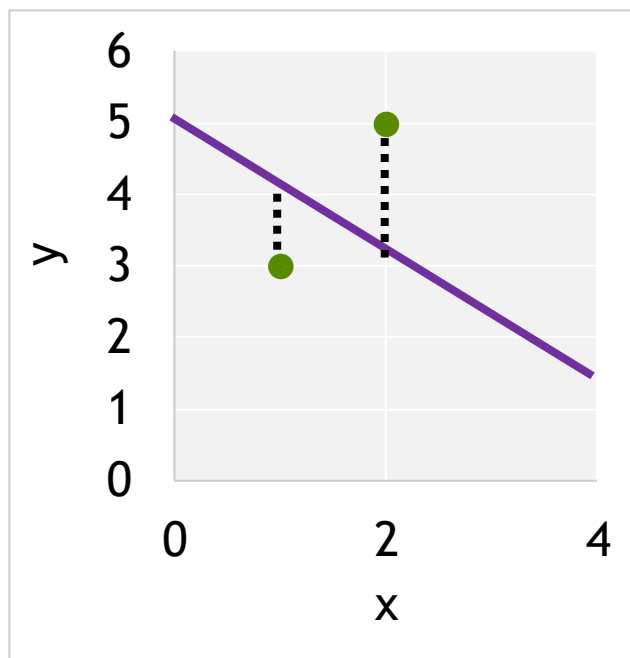
$$m = -1$$

$$b = 5$$

더욱 단순한 모델

$$y = mx + b$$

x	y	$\hat{y}$	err^2
1	3	4	1
2	5	3	4
MSE =			2.5
RMSE =			1.6



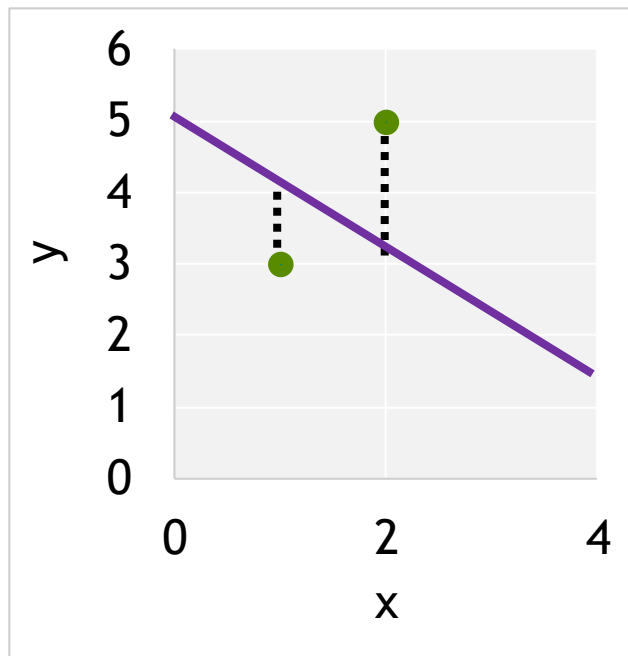
$$MSE = \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2}$$

더욱 단순한 모델

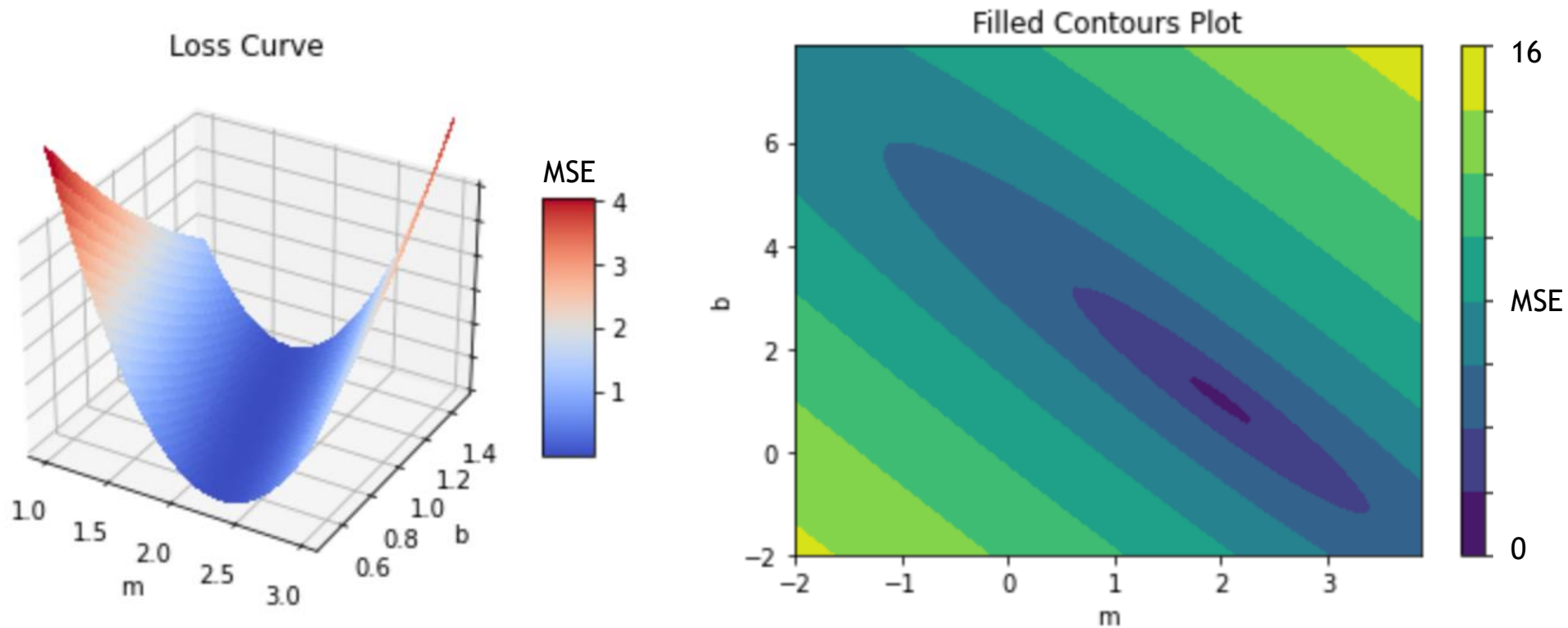
$$y = mx + b$$

x	y	$\hat{y}$	err^2
1	3	4	1
2	5	3	4
MSE =			2.5
RMSE =			1.6

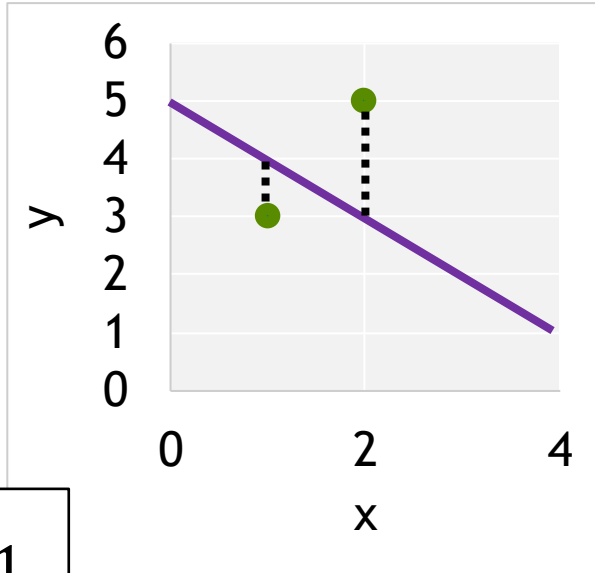


```
1 data = [(1, 3), (2, 5)]
2 m = -1
3 b = 5
4
5 def get_mse(data, m, b):
6     """Calculates Mean Square Error"""
7     n = len(data)
8     squared_error = 0
9     for x, y in data:
10         # Find predicted y
11         y_hat = m*x+b
12         # Square difference between
13         # prediction and true value
14         squared_error += (
15             y - y_hat)**2
16     # Get average squared difference
17     mse = squared_error / n
18     return mse
19
```

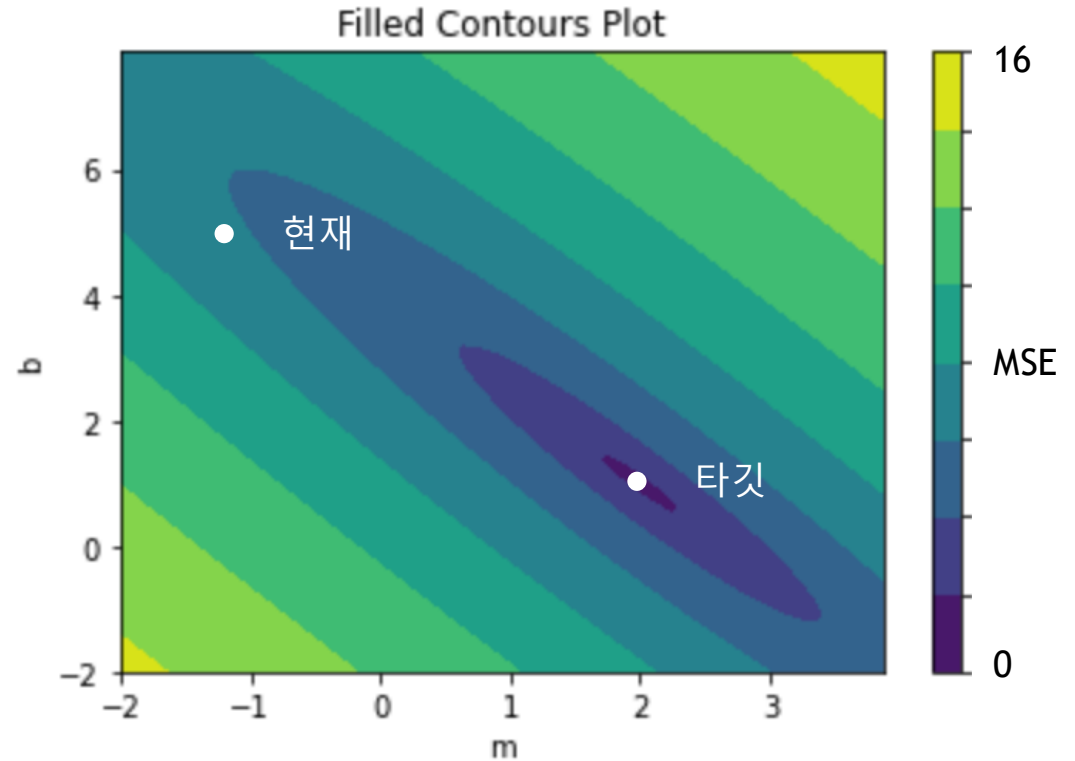
손실 곡선 (THE LOSS CURVE)



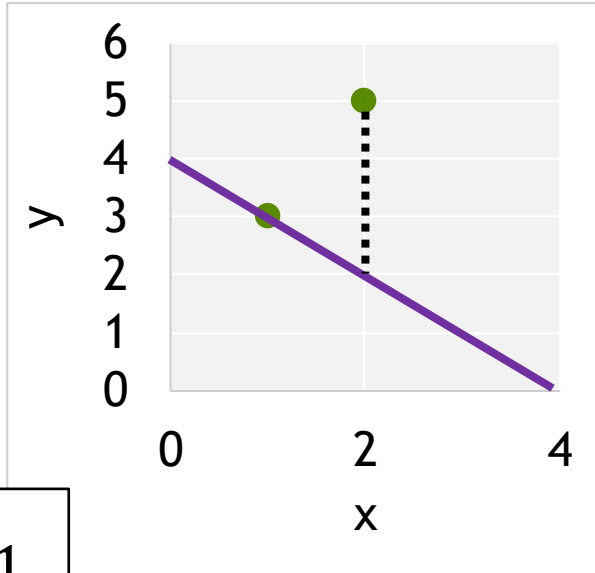
손실 곡선 (THE LOSS CURVE)



$$m = -1$$
$$b = 5$$

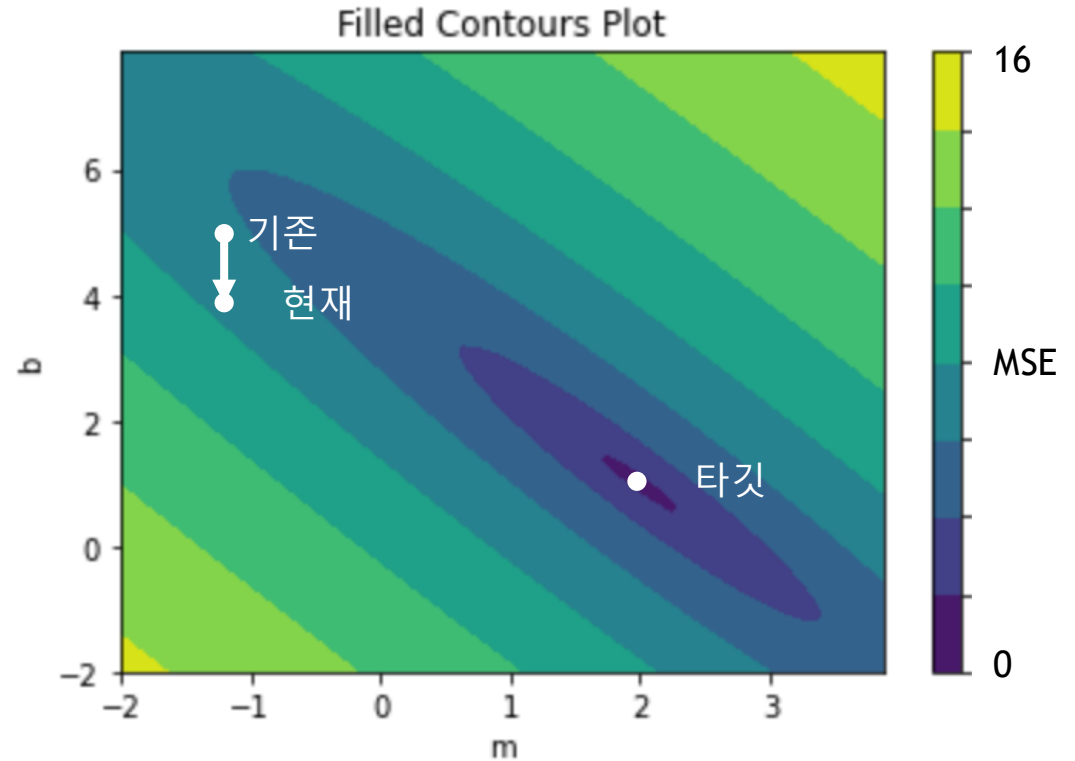


손실 곡선 (THE LOSS CURVE)

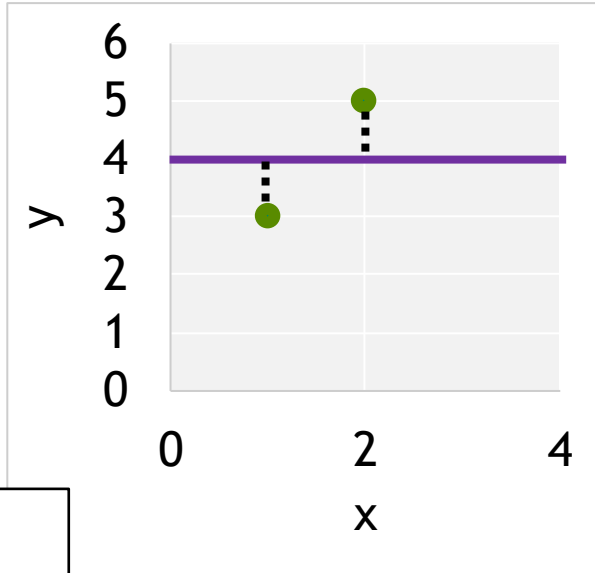


$$m = -1$$

$$b = 4$$

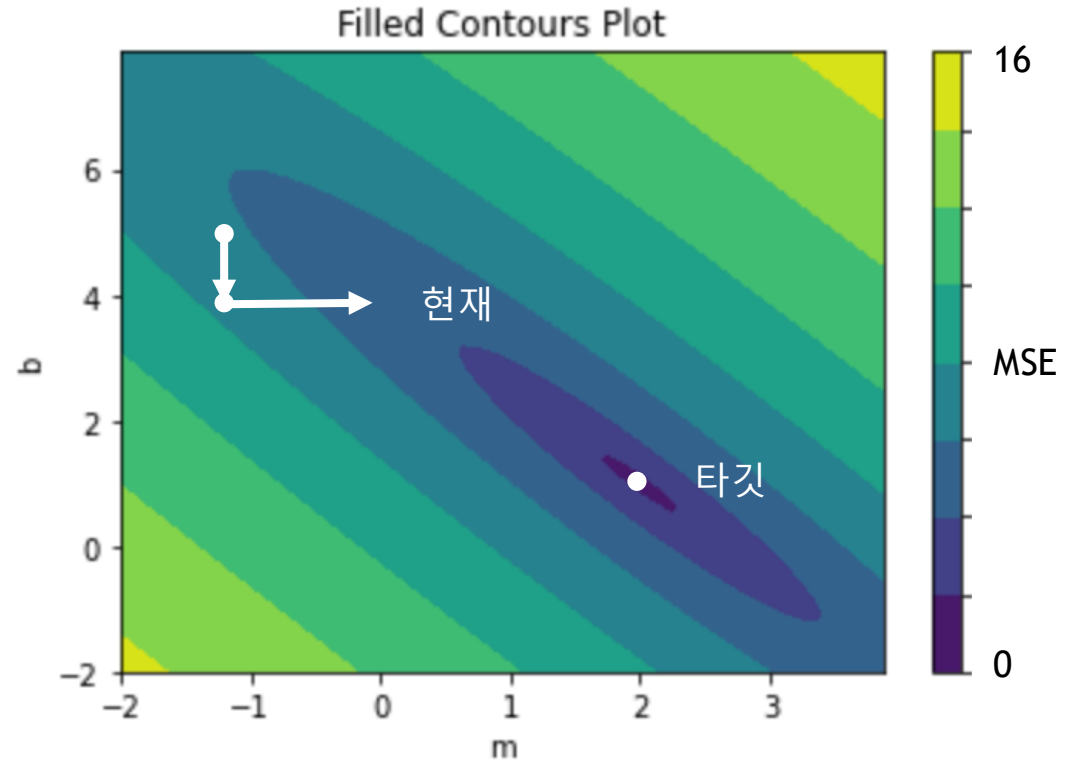


손실 곡선 (THE LOSS CURVE)



$$m = 0$$

$$b = 4$$



손실 곡선 (THE LOSS CURVE)

그라디언트
(Gradient)

어떤 방향으로 손실이 가장 많이 감소하는가

λ : 학습률
(Learning rate)

이동 거리

에포크
(Epoch)

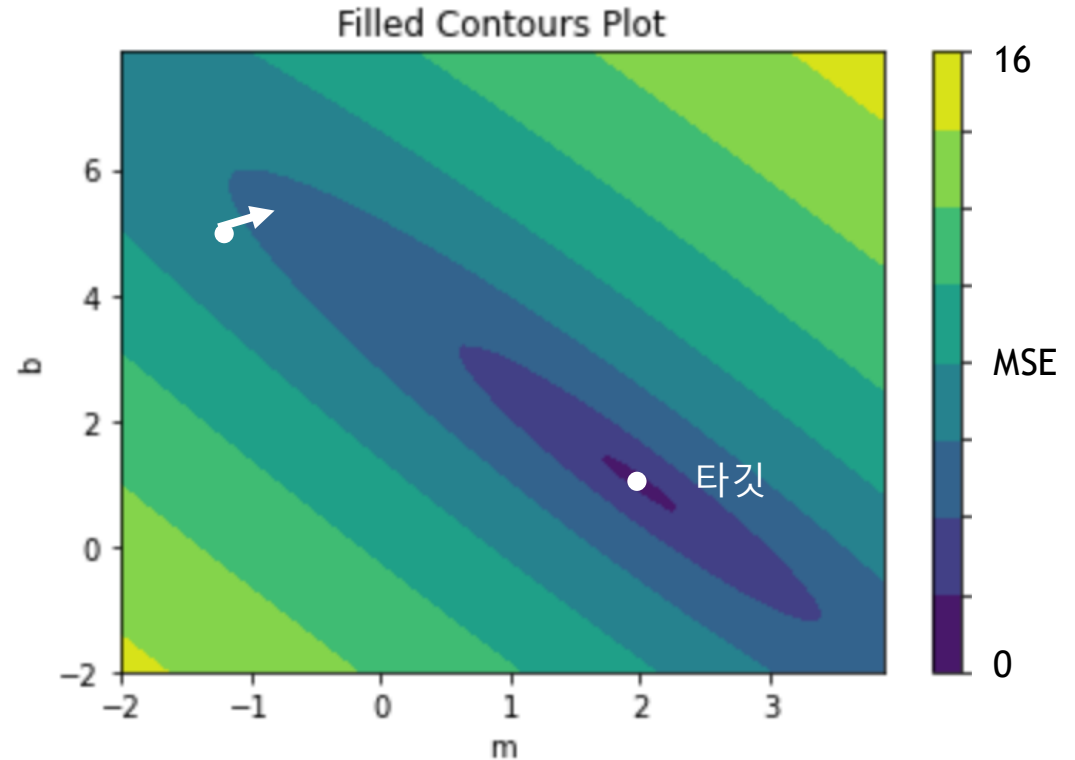
전체 데이터셋으로 모델 업데이트

배치 (Batch)

전체 데이터셋 중 특정 샘플

단계 (Step)

가중치 매개변수에 대한 업데이트



손실 곡선 (THE LOSS CURVE)

그라디언트
(Gradient)

어떤 방향으로 손실이 가장 많이
감소하는가

λ : 학습률
(Learning
rate)

이동 거리

에포크
(Epoch)

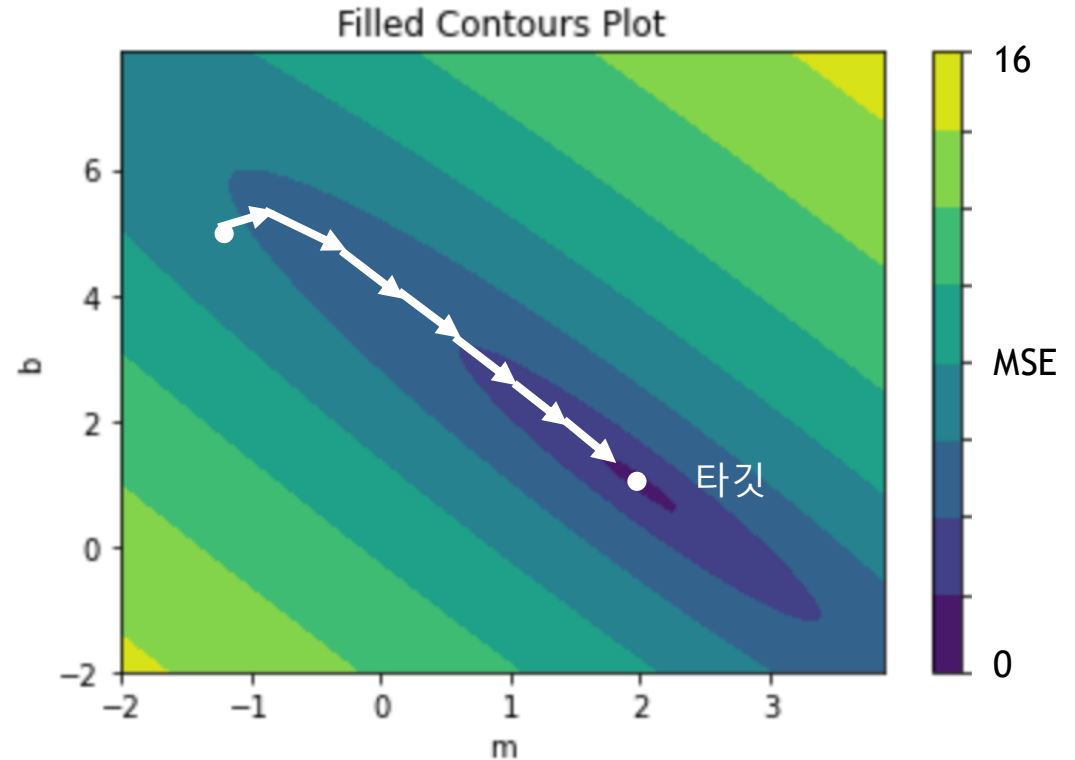
전체 데이터셋으로 모델 업데이트

배치 (Batch)

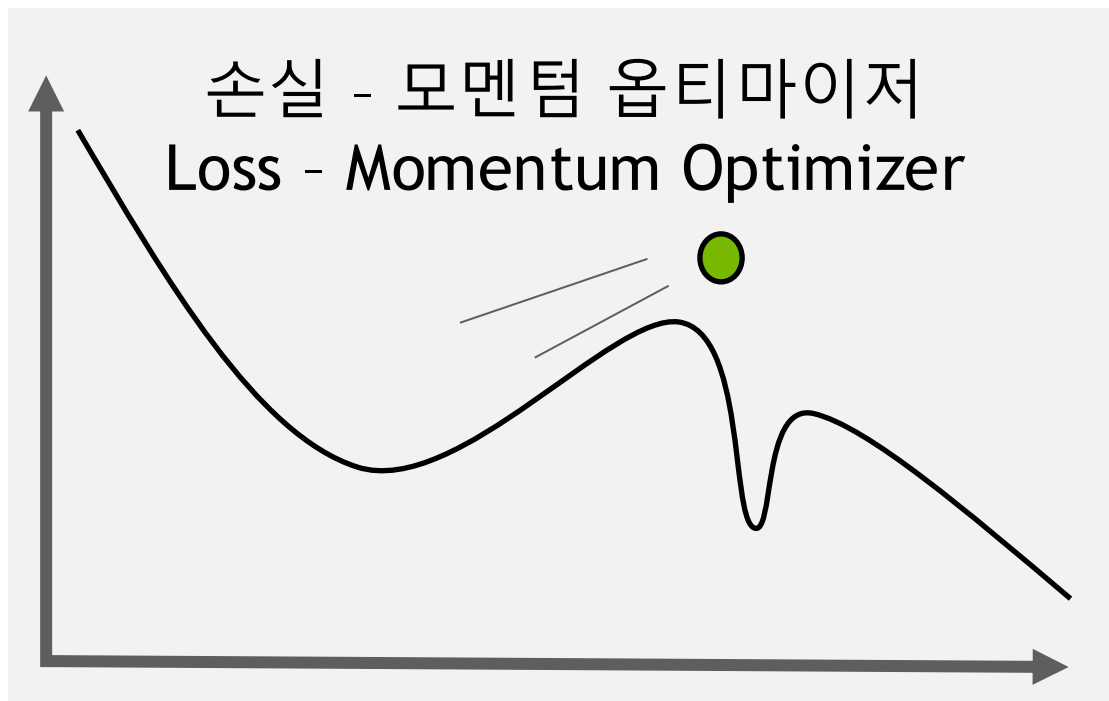
전체 데이터셋 샘플

단계 (Step)

가중치 매개변수에 대한 업데이트



옵티마이저 (OPTIMIZERS)

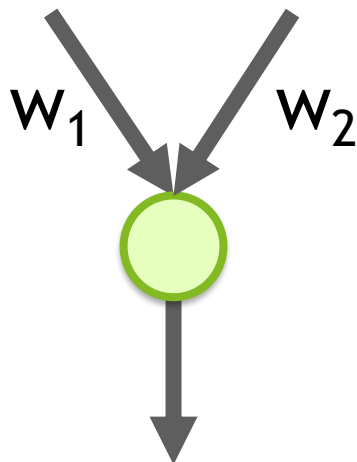


- Adam
- Adagrad
- RMSprop
- SGD



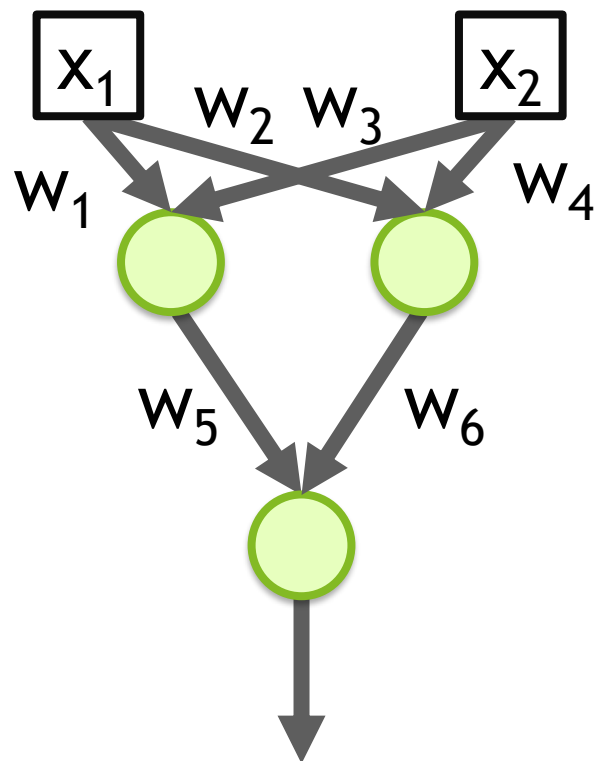
뉴런에서 네트워크로

네트워크 구축



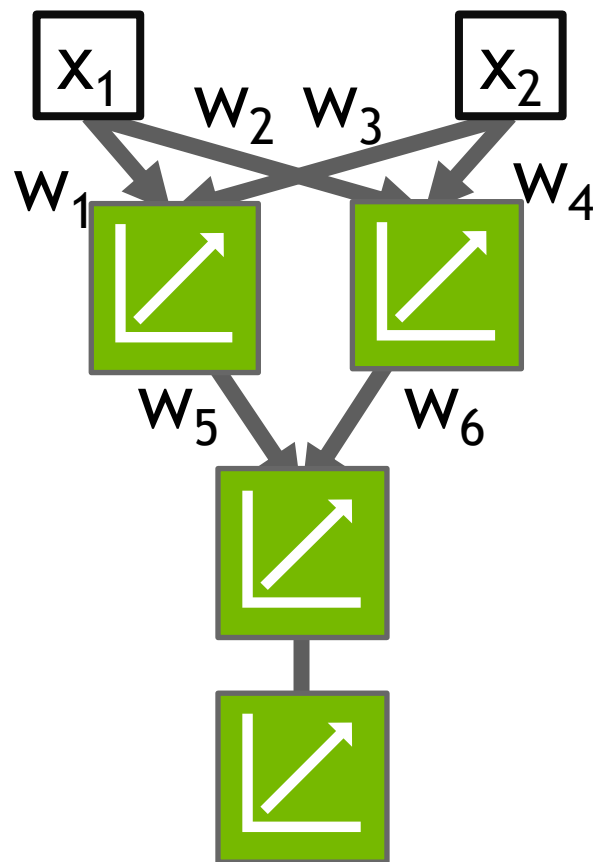
- 더 많은 변수로의 확장

네트워크 구축



- 더 많은 변수로의 확장
- 뉴런들간의 연결

네트워크 구축



- 더 많은 변수로의 확장
- 뉴런들간의 연결
- 모든 회귀가 선형이면 출력 결과도 선형 회귀



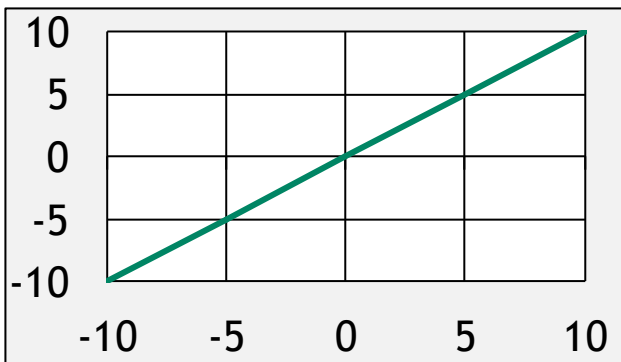
활성화 함수 ACTIVATION FUNCTIONS

활성화 함수 (ACTIVATION FUNCTIONS)

Linear(선형)

$$\hat{y} = wx + b$$

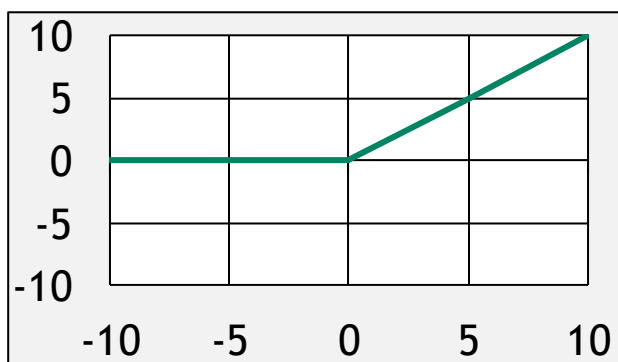
```
1 # Multiply each input
2 # with a weight (w) and
3 # add intercept (b)
4 y_hat = wx+b
```



ReLU

$$\hat{y} = \begin{cases} wx + b & \text{if } wx + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

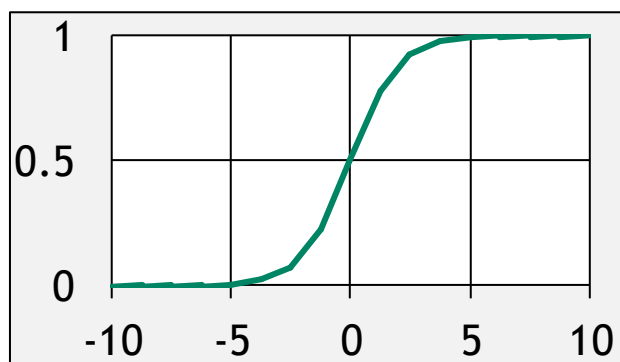
```
1 # Only return result
2 # if total is positive
3 linear = wx+b
4 y_hat = linear * (linear > 0)
```



Sigmoid(시그모이드)

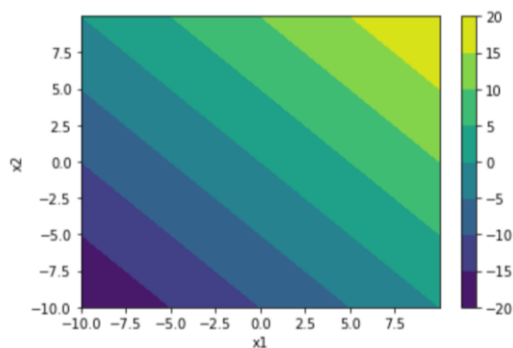
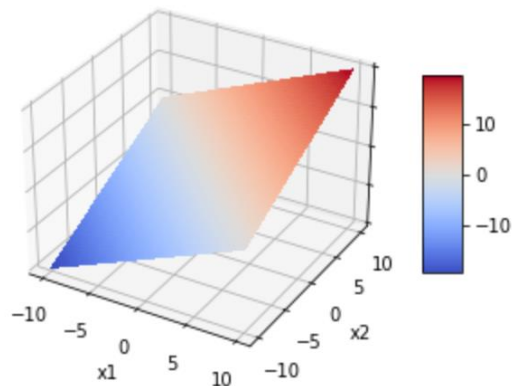
$$\hat{y} = \frac{1}{1 + e^{-(wx+b)}}$$

```
1 # Start with line
2 linear = wx + b
3 # Warp to - inf to 0
4 inf_to_zero = np.exp(-1 * linear)
5 # Squish to -1 to 1
6 y_hat = 1 / (1 + inf_to_zero)
```

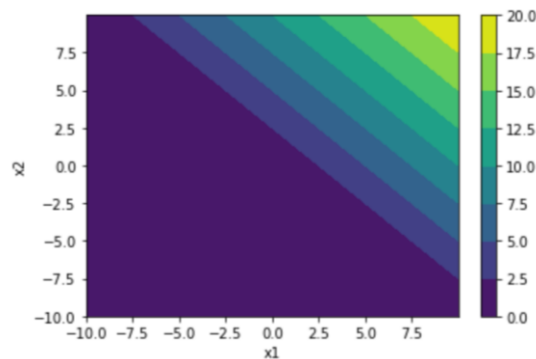
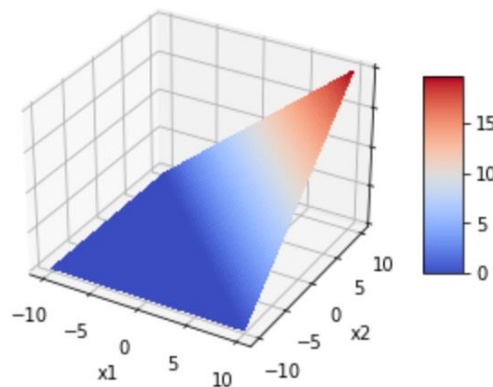


활성화 함수 (ACTIVATION FUNCTIONS)

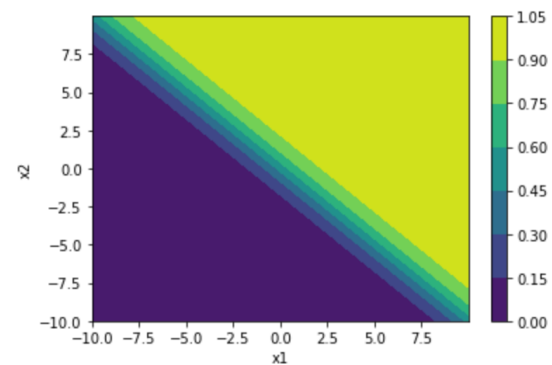
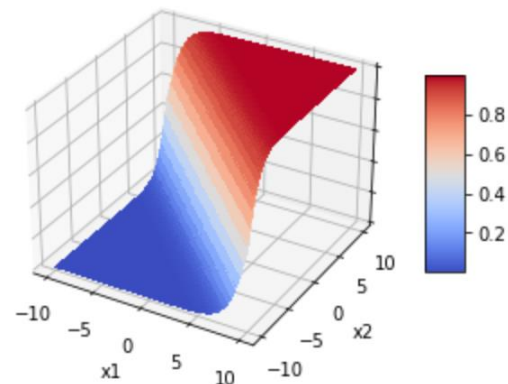
Linear(선형)



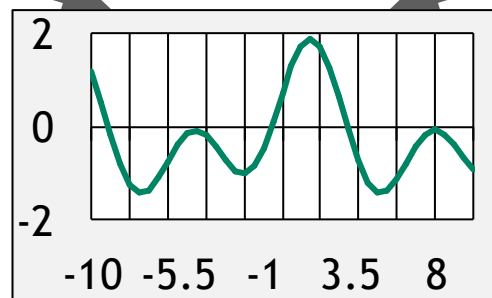
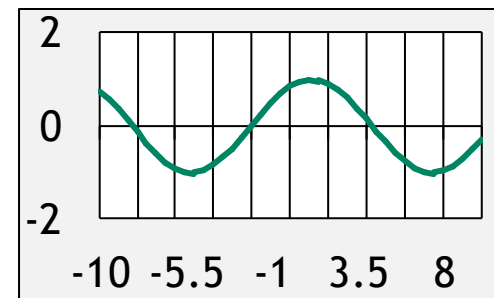
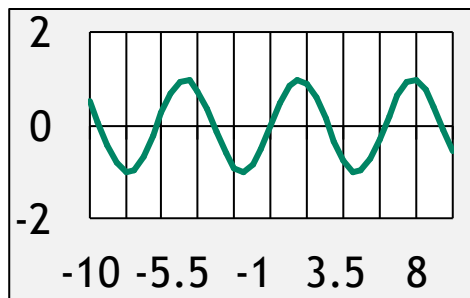
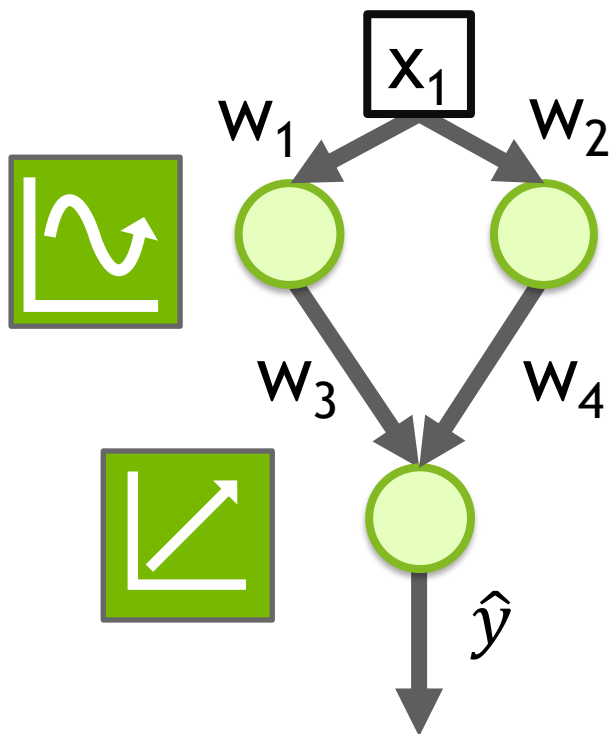
ReLU



Sigmoid(시그모이드)



활성화 함수 (ACTIVATION FUNCTIONS)





과적합
OVERFITTING

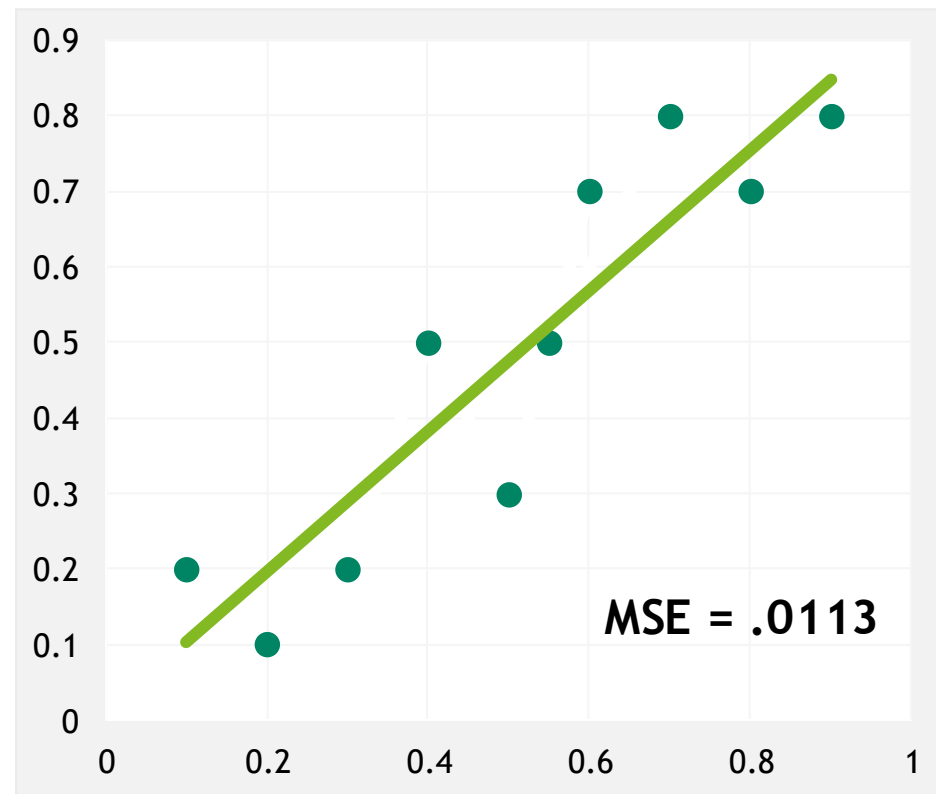
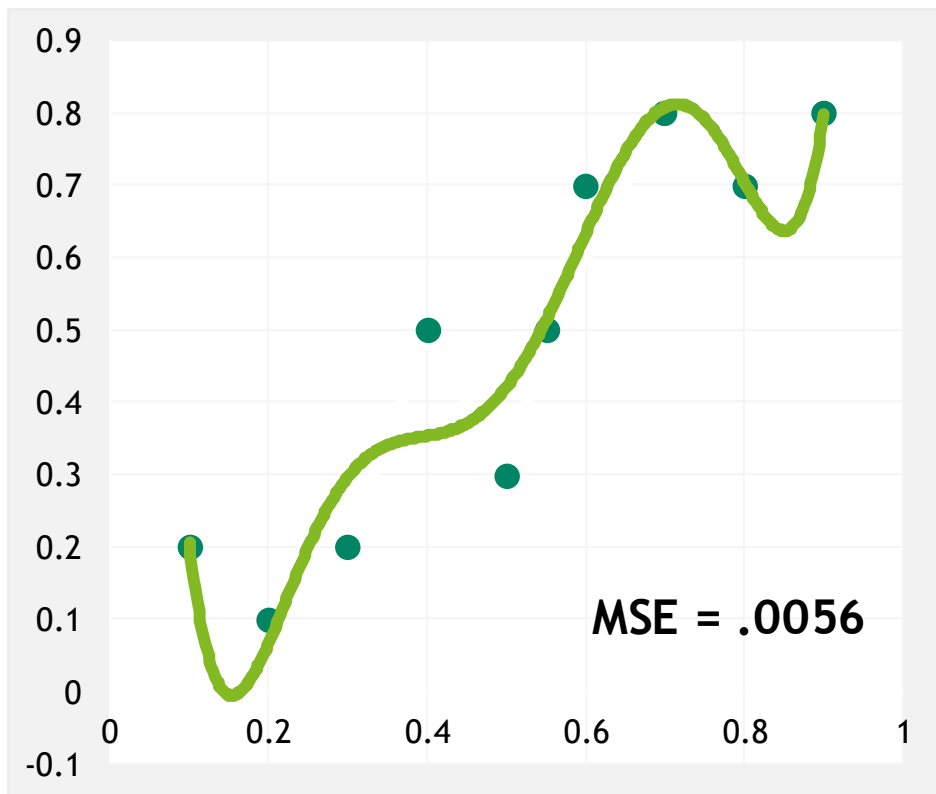
과적합 (OVERFITTING)

거대한 뉴럴 네트워크를 구축하면 되지 않을까?



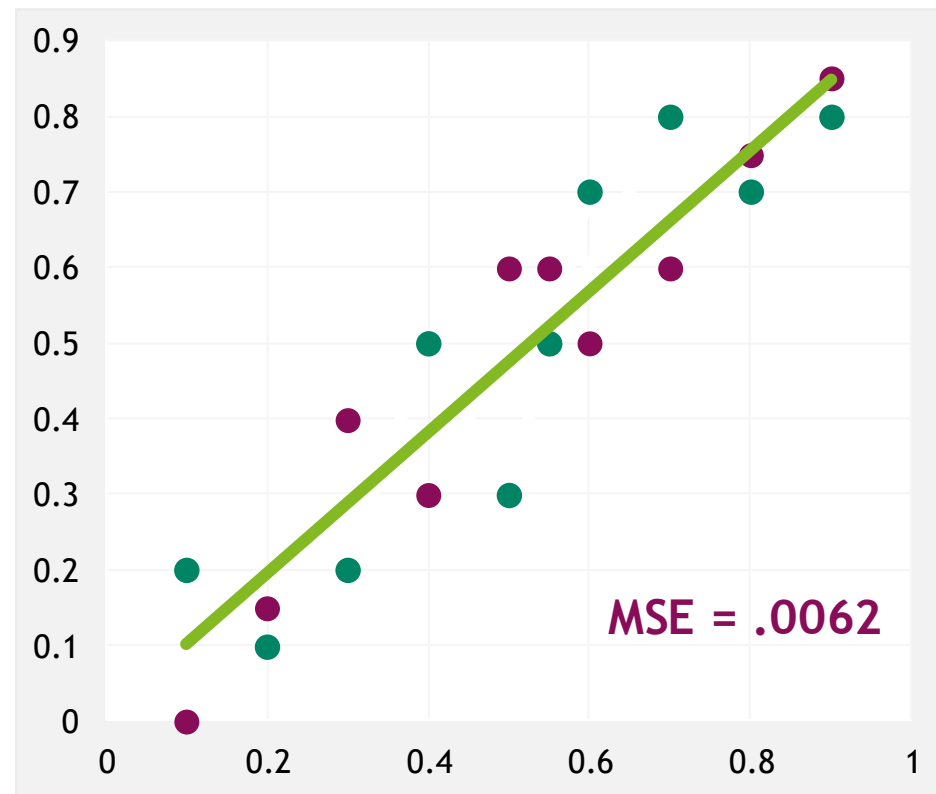
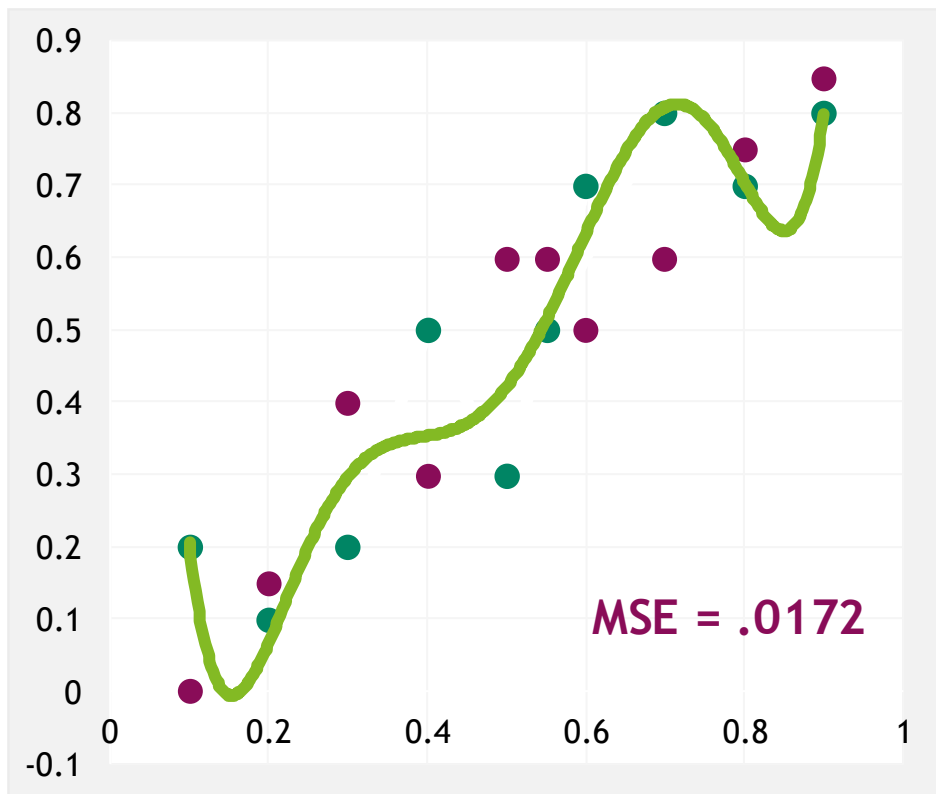
과적합 (OVERFITTING)

어떤 추세선이 더 나은가?



과적합 (OVERFITTING)

어떤 추세선이 더 나은가?



트레이닝 데이터와 검증 데이터 비교

암기 회피(Avoid memorization)

트레이닝 데이터 (Training data)

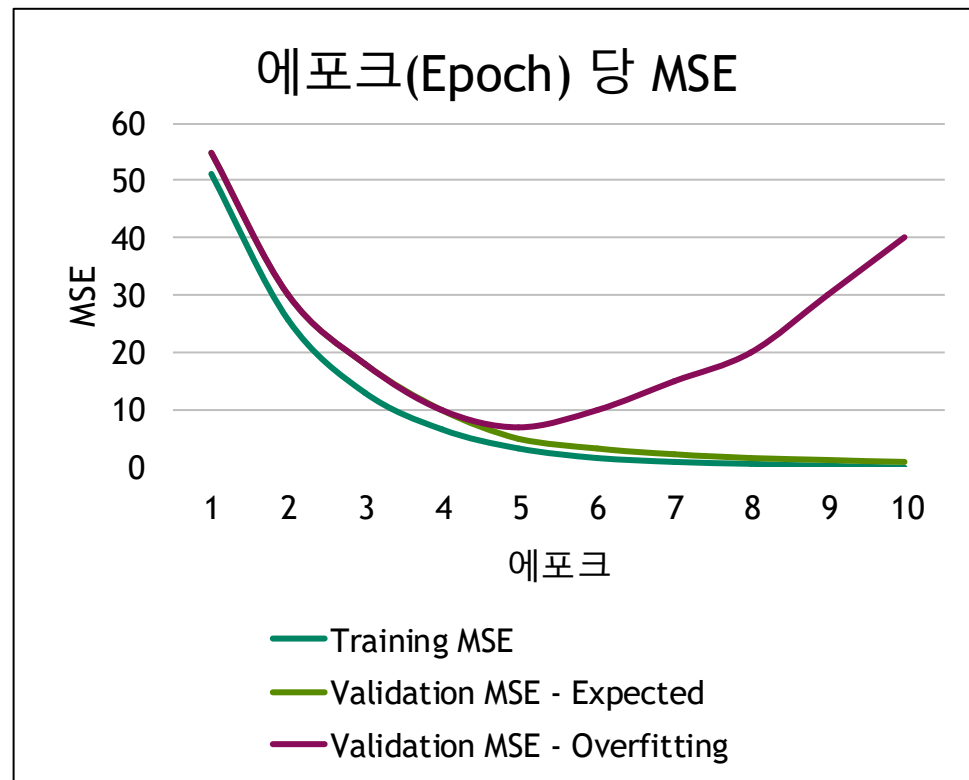
- 학습해야 하는 모델의 핵심 데이터세트

검증 데이터 (Validation data)

- 정말로 이해하는지(일반화 가능한지)를 확인하기 위한 모델의 새 데이터

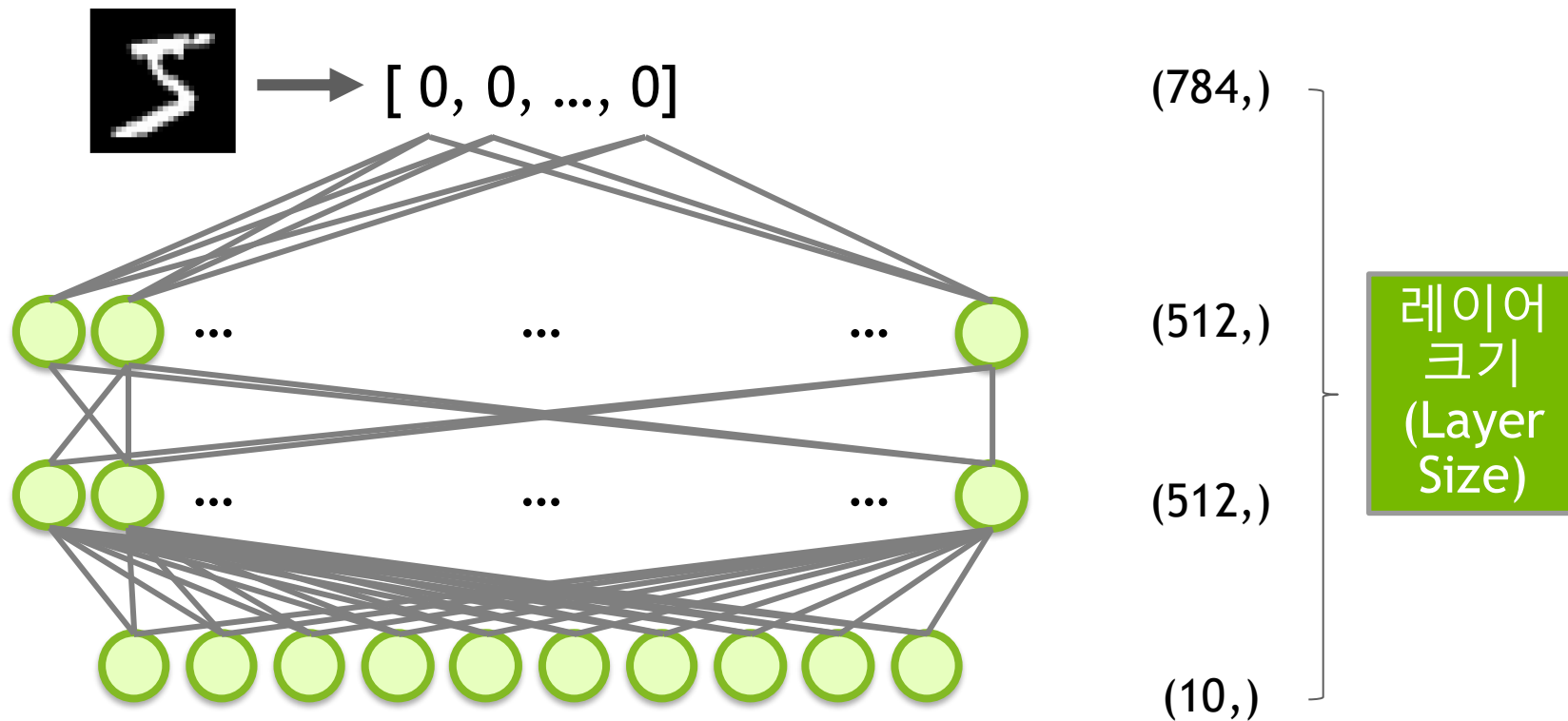
과적합 (Overfitting)

- 모델이 트레이닝 데이터에 대한 좋은 성능을 보이지만 검증 데이터는 그렇지 못함(암기의 증거)
- 정확도 및 손실이 두 데이터세트 간에 유사해야 이상적임

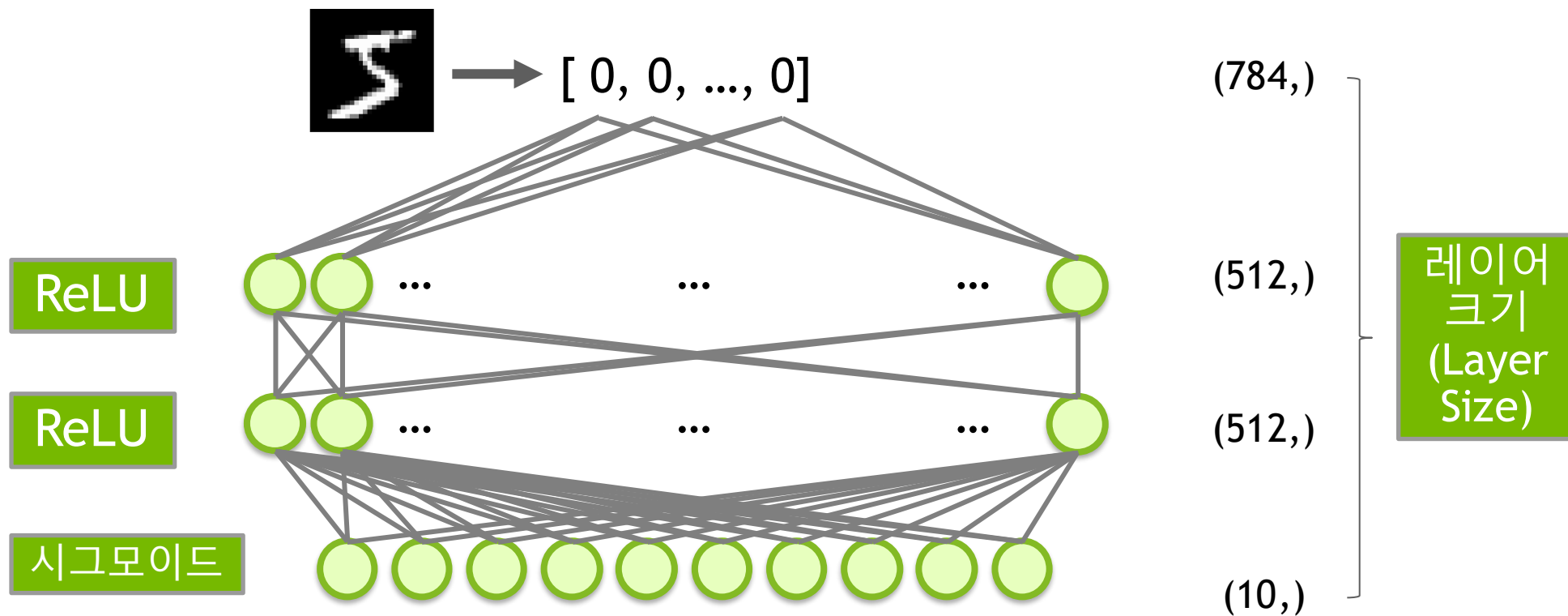


회귀에서 분류로
FROM REGRESSION TO
CLASSIFICATION

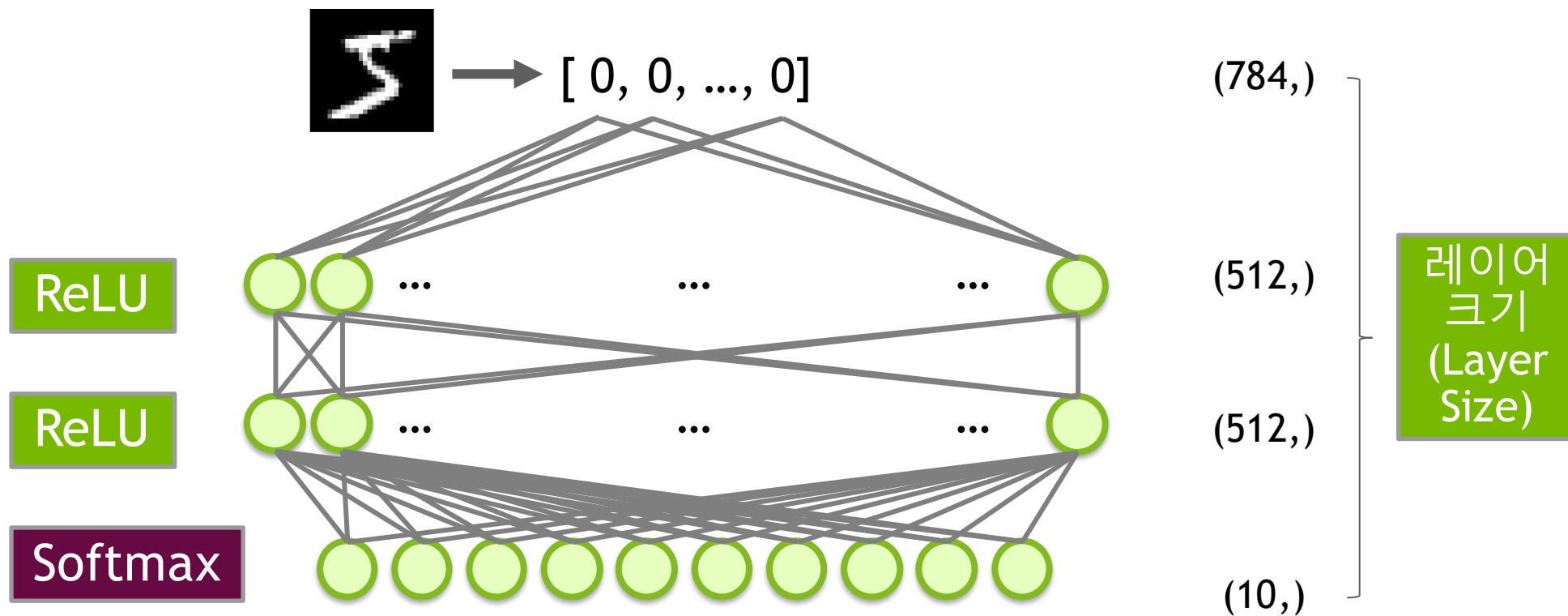
MNIST 모델



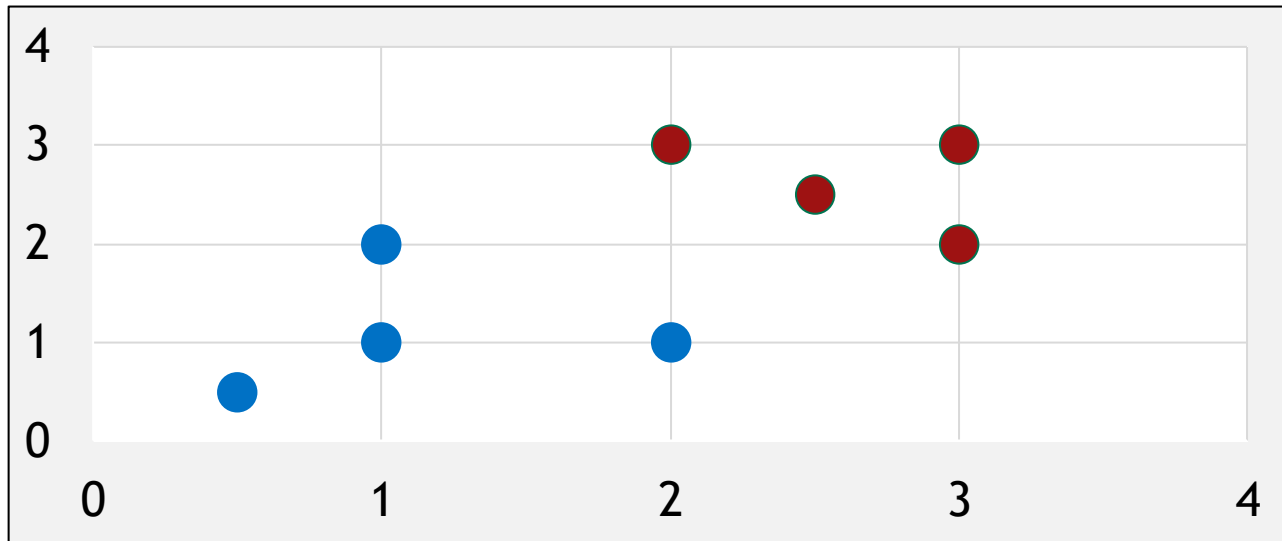
MNIST 모델



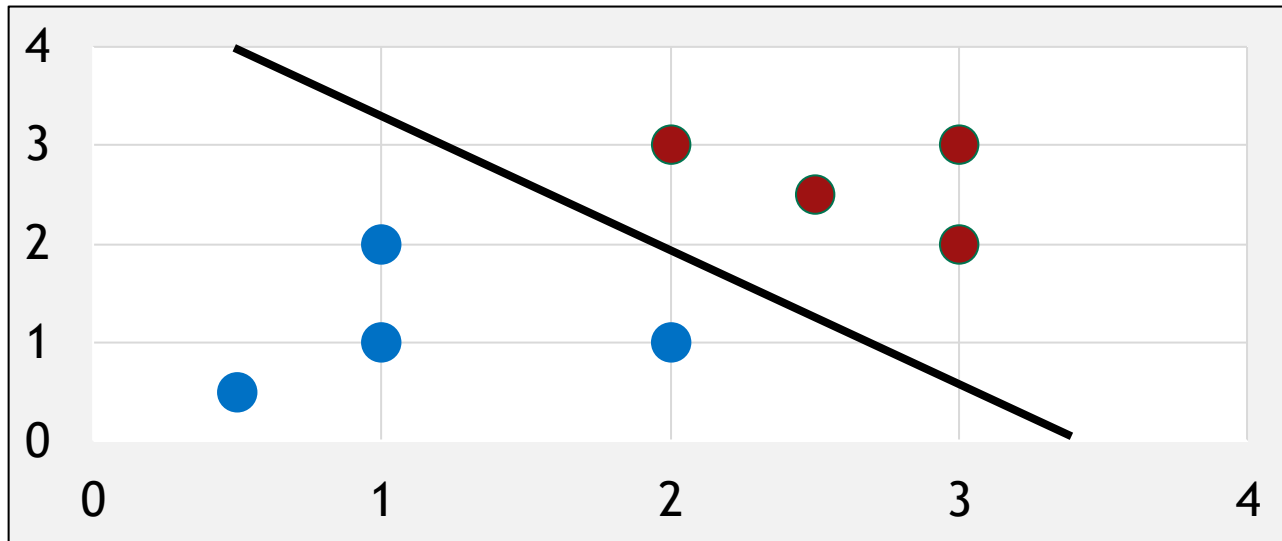
MNIST 모델



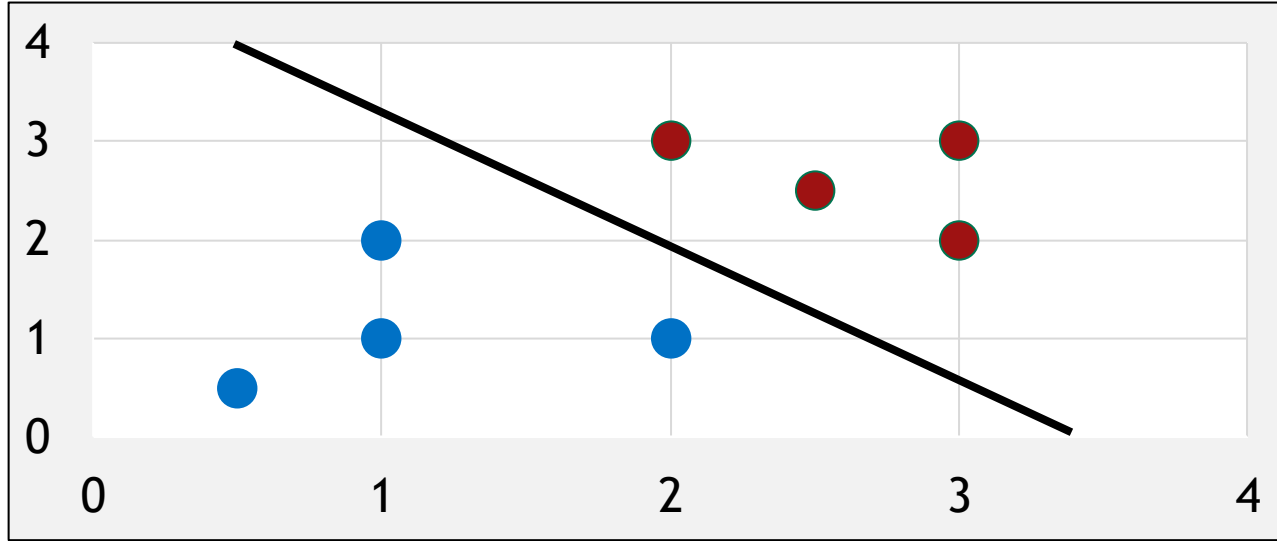
RMSE로 확률 예측?



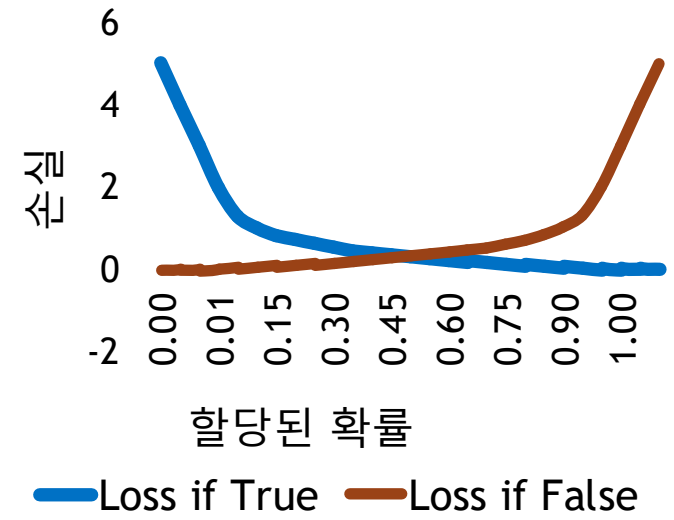
RMSE로 확률 예측?



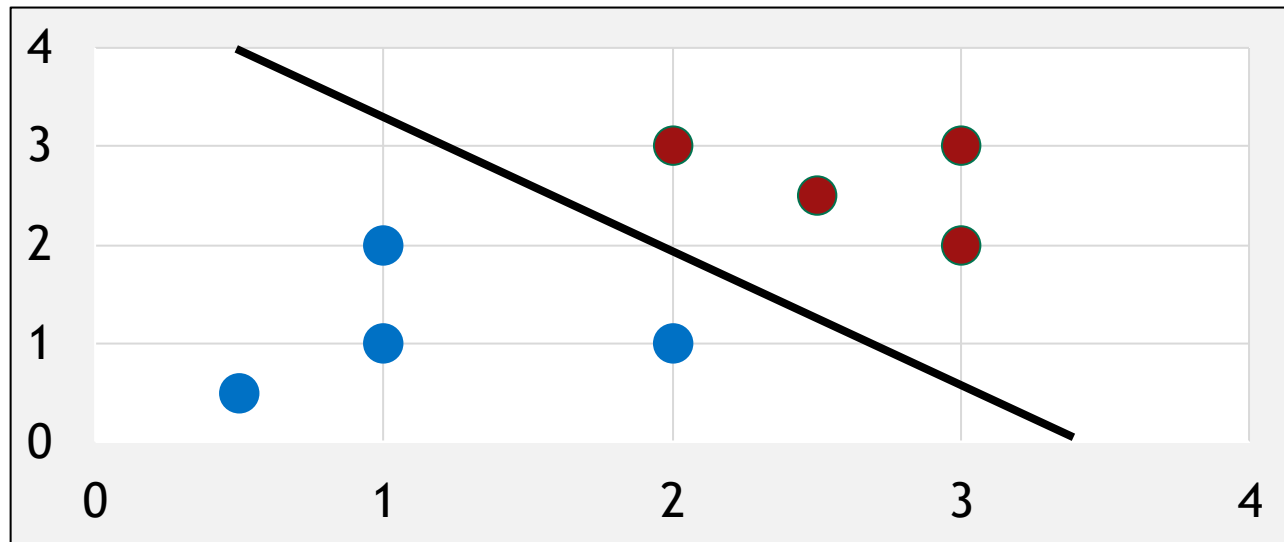
크로스 엔트로피 (CROSS ENTROPY)



크로스 엔트로피
(CROSS ENTROPY)
파란색 점 예측



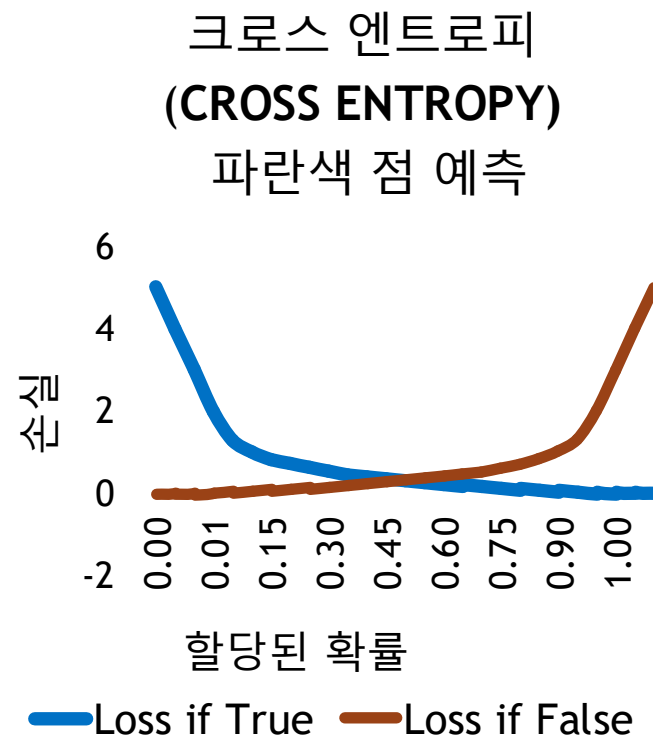
크로스 엔트로피 (CROSS ENTROPY)



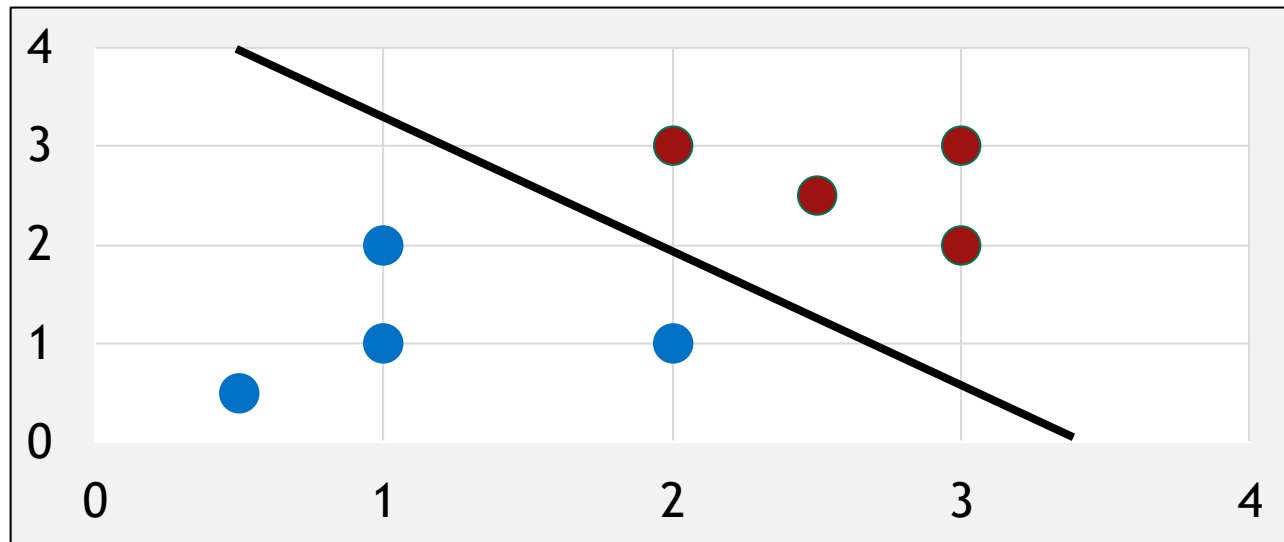
$$Loss = -(t(x) \cdot \log(p(x)) + (1 - t(x)) \cdot \log(1 - p(x)))$$

$t(x) = target$ (1 if True, 0 if False)

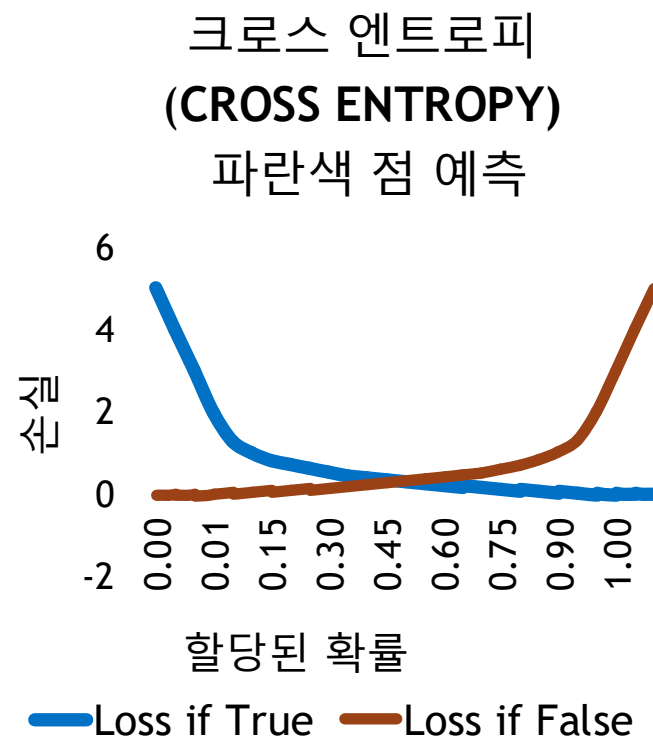
$p(x) = probability prediction of point x$



크로스 엔트로피 (CROSS ENTROPY)



```
1 def cross_entropy(y_hat, y_actual):  
2     """Infinite error for misplaced confidence."""  
3     loss = log(y_hat) if y_actual else log(1-y_hat)  
4     return -1*loss
```

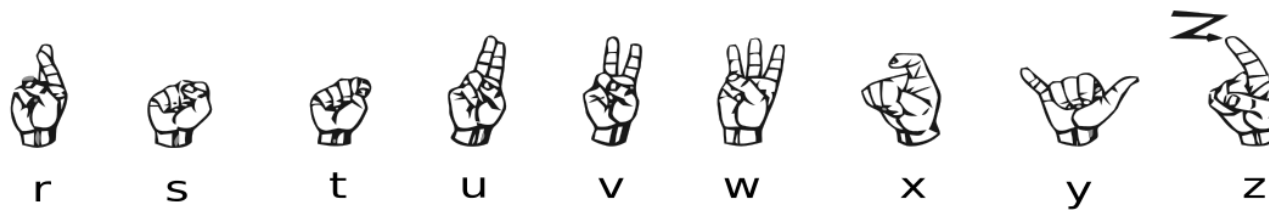
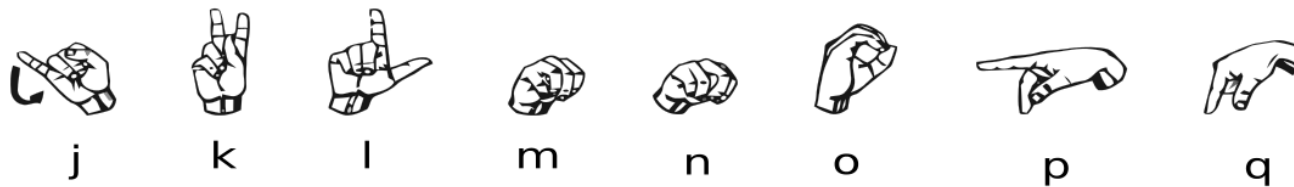
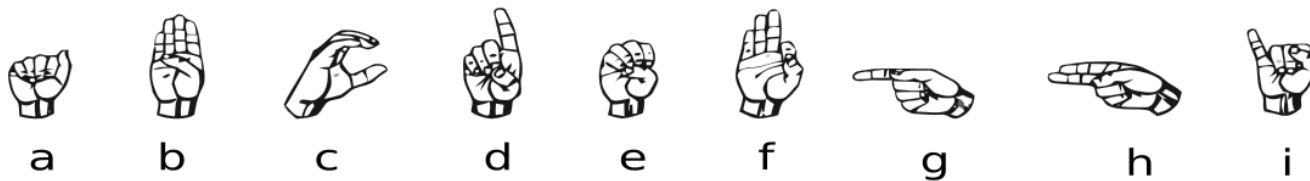




종합 정리

연습 문제

미국 수화 알파벳





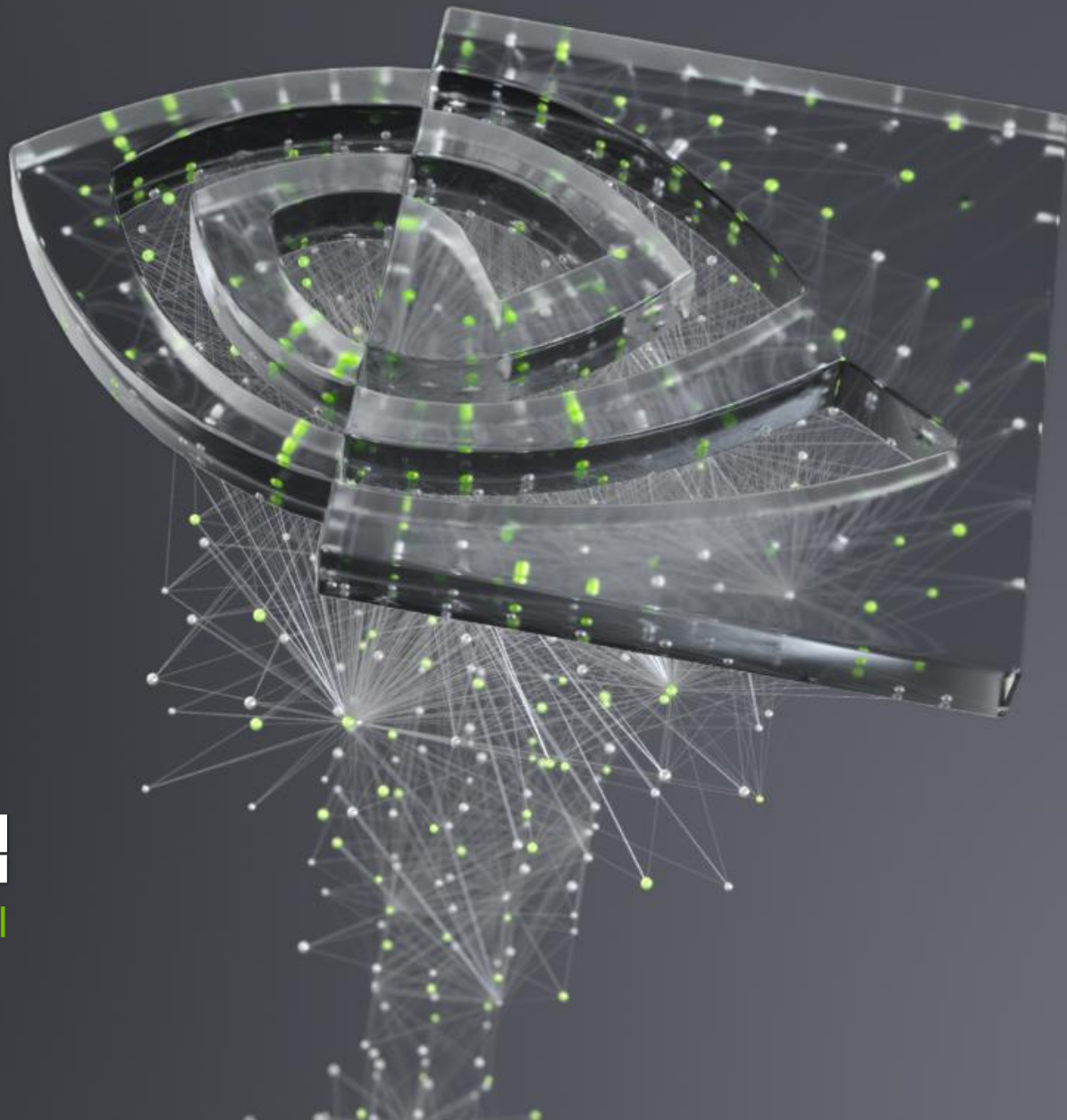
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부록: 그래디언트 하강법

컴퓨터가 미적분에 대한 편법을 사용할 수 있도록 돕기



에러를 통한 학습

$$MSE = \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2 = \frac{1}{n} \sum_{i=1}^n (y - (mx + b))^2$$

$$MSE = \frac{1}{2} ((3 - (m(1) + b))^2 + (5 - (m(2) + b))^2)$$

$$\frac{\partial MSE}{\partial m} = 9m + 5b - 23$$

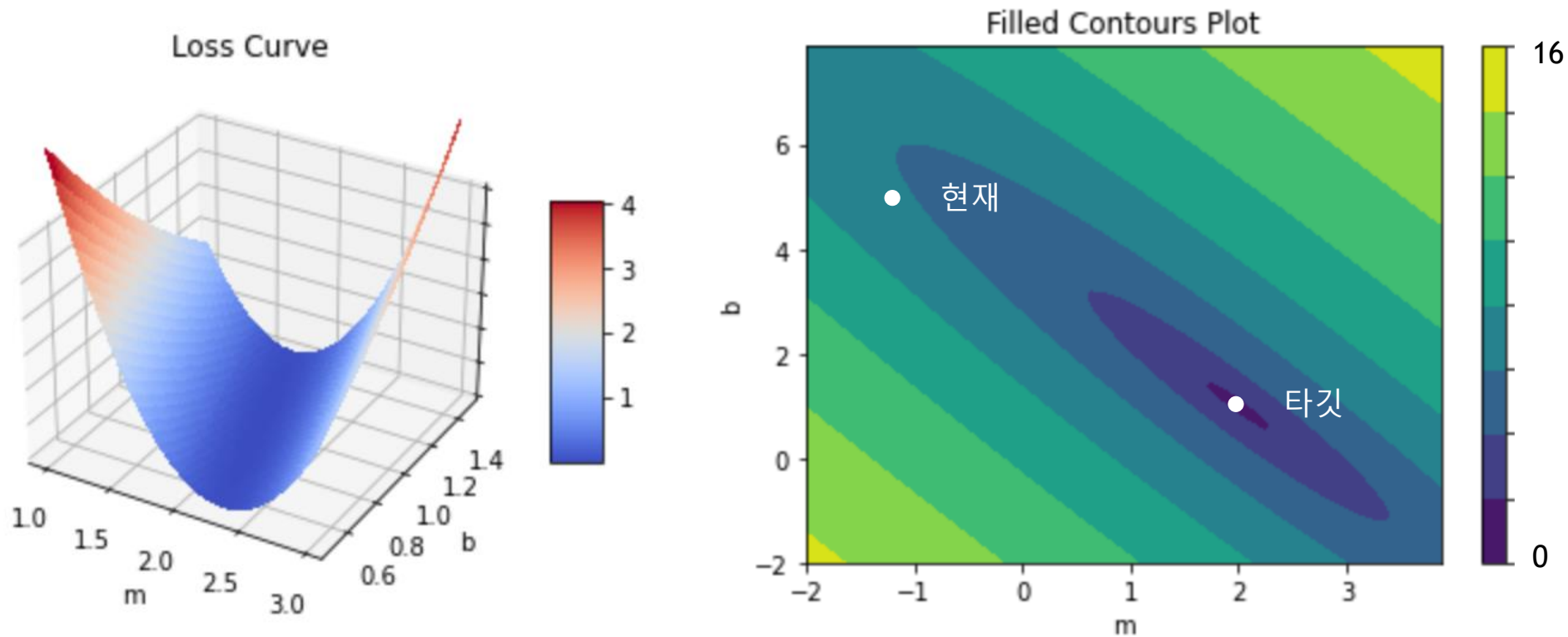
$$\frac{\partial MSE}{\partial b} = 5m + 3b - 13$$

$$\frac{\partial MSE}{\partial m} = -7$$

$$\frac{\partial MSE}{\partial b} = -3$$

$$m = -1$$
$$b = 5$$

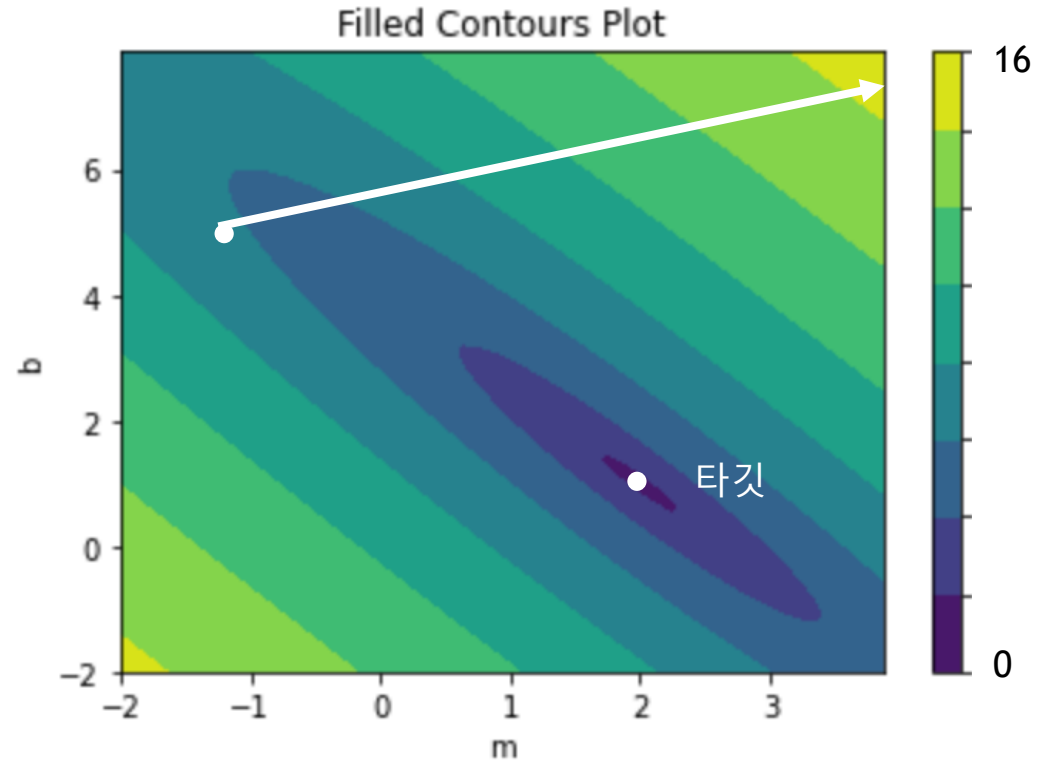
손실 곡선 (THE LOSS CURVE)



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$$\frac{\partial MSE}{\partial m} = -7$$

$$\frac{\partial MSE}{\partial b} = -3$$

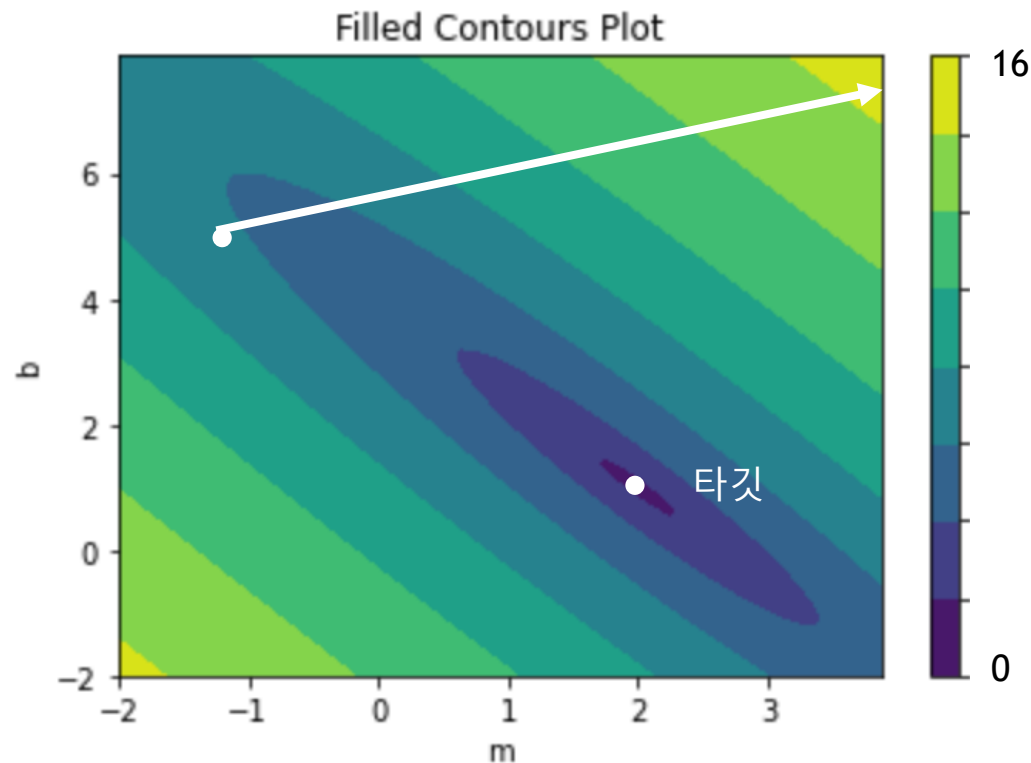


손실 곡선 (THE LOSS CURVE)

$$\frac{\partial MSE}{\partial m} = -7 \quad \frac{\partial MSE}{\partial b} = -3$$

$$m := m - \lambda \frac{\partial MSE}{\partial m}$$

$$b := b - \lambda \frac{\partial MSE}{\partial b}$$

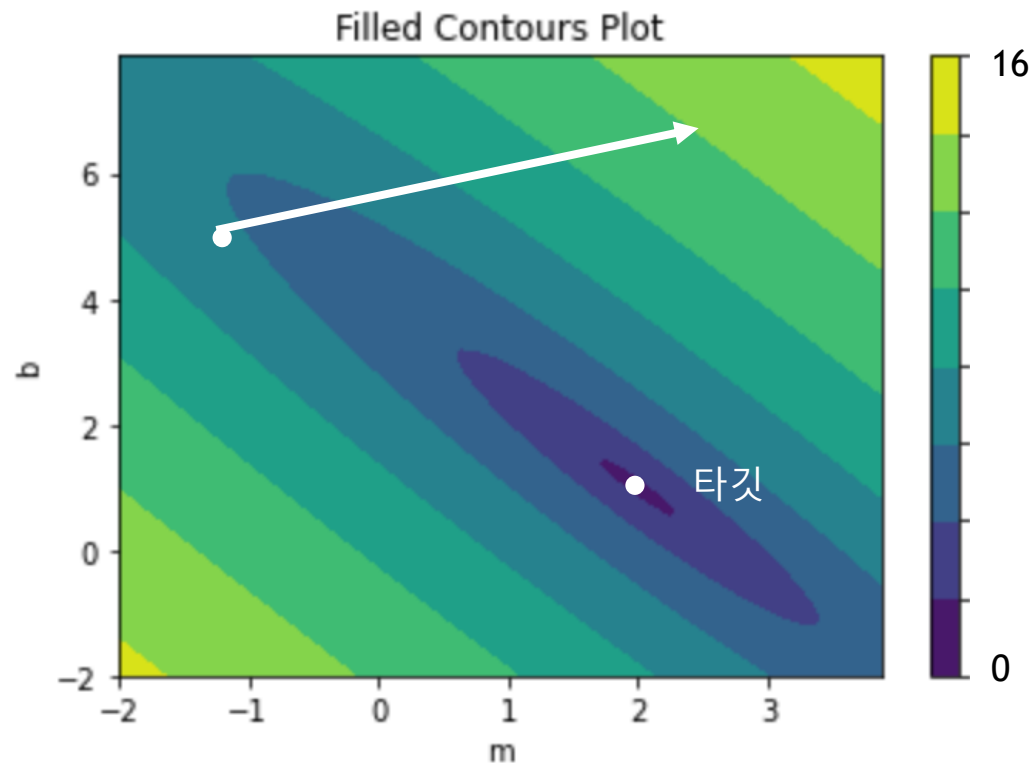


손실 곡선 (THE LOSS CURVE)

$$\frac{\partial MSE}{\partial m} = -7 \quad \frac{\partial MSE}{\partial b} = -3$$

$$m := m - \lambda \frac{\partial MSE}{\partial m} \quad \lambda = .5$$

$$b := b - \lambda \frac{\partial MSE}{\partial b}$$



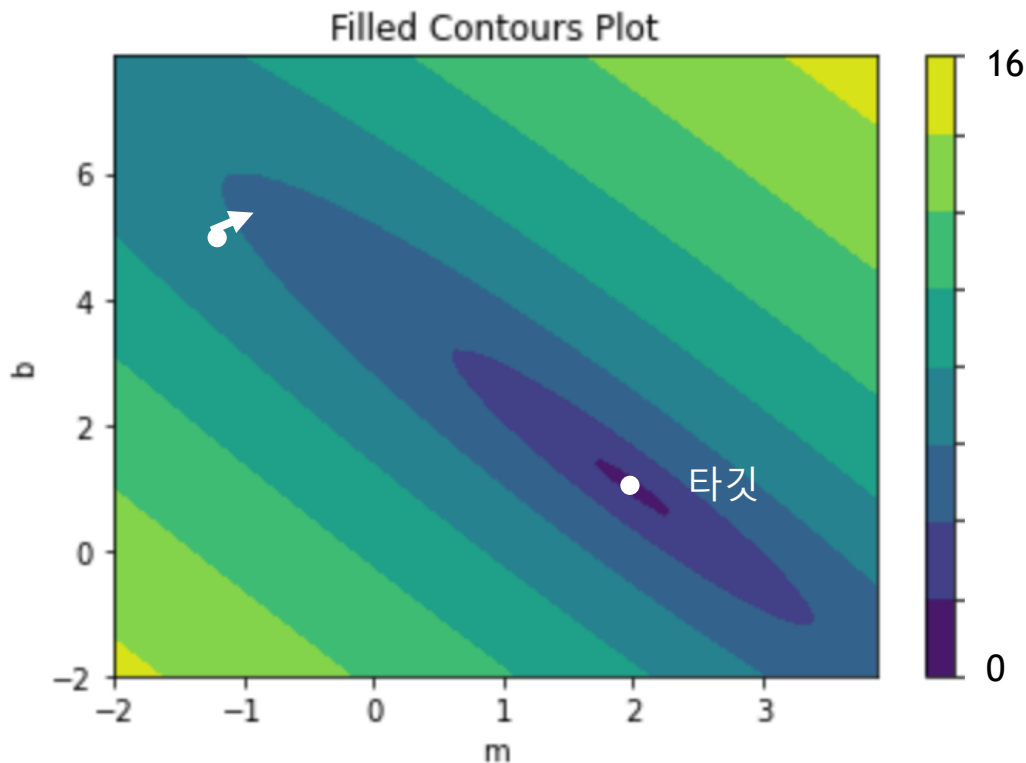
손실 곡선 (THE LOSS CURVE)

$$\frac{\partial MSE}{\partial m} = -7 \quad \frac{\partial MSE}{\partial b} = -3$$

$$m := m - \lambda \frac{\partial MSE}{\partial m}$$

$$\lambda = .005$$

$$b := b - \lambda \frac{\partial MSE}{\partial b}$$



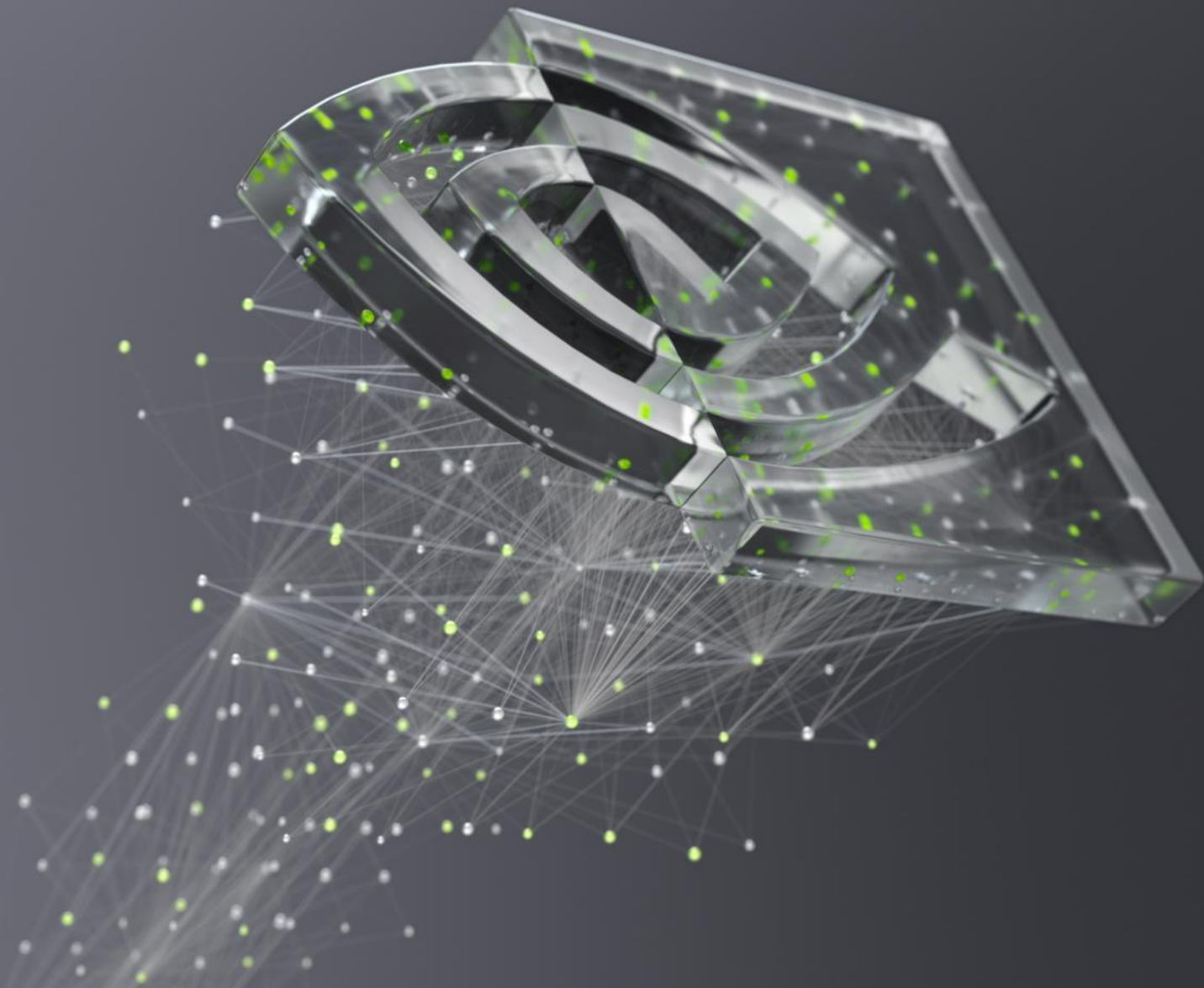
손실 곡선 (THE LOSS CURVE)

$$\lambda = .1$$

$$m := -1 - 7\lambda = -1.7$$

$$b := 5 - 3\lambda = 5.3$$





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